Mapping of Human Hand Dynamics onto a Robotic Hand for Dexterous Telemanipulation

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Abstract

In this study, some of the problems of telemanipulation systems with multibody teleoperator mechanisms are addressed. In our specific setup, telemanipulation occurs by means of a robotic hand having 3 fingers. Although having functional similarities, the robotic hand used is not a anthropomorphic one in its dimensions and articulations. The differences between the governing human operator hand kinematics and the teleoperator kinematics exclude a straigthforward bijective mapping between the two. A mapping using state space order reduction of the human hand joint space with help of physiological knowledge is proposed. Out of the reduced state space of the human hand, necessary control commands for the robotic controller will be derived.

Zusammenfassung

In dieser Arbeit sind einige Probleme vom Telemanipulationssystemen mit Mehrkörpermechanismen dargestellt. Im betrachteten System, die Telemanipulation erfolgt mitHilfe einer Roboterhand mit 3 Fingern. Obwohl diese funktionelle Ähnlichkeiten zur menschlichen Hand hat, ist die Roboterhand keine anthropomorphische Kopie im Bezug aif Abmessung und Artikulation. Die bedeutsamen Unterschiede zwischen Roboter- und menschlicher Hand schliessen eine einfache bijektive Abbildung aus. Hier wurde eine Abbildung basierend auf einer Zustandsraum Ordnungsreduktion anhand physiologischen Erkenntnissen vorgeschlagen. Ausgehend von dem reduzierten Zustandsraum der menschlichen Hand ausgehend werden die benötigten Stellgrößen für die Roboterhand abgeleitet.

Çok zaman ve emek harcadığınız bir çalışmanın son cümlelerini yazmak hem rahatlatıcı, hem de düşündürücü. Binlerce yıldır yaşayan insanoğlu hala aradığı huzurun, mutluluğun formülüne ulaşamamış, bugün hala dünyanın birçok yerinde insan insanı öldürürken, insanoğlunun gelişmişliğinden bahsediliyor; halbuki insan en temel sorununu çözememiş, paylaşmayı beceremiyor. Bunu çözemeyen insana sorulacak soru ise şu olabilir: ”Üretilen tüm yeniliklerin kötüye kullanıldığı bir dünyada yeniyi aramak niye?”

Bu tezi hazırlarken yıllardır aklımda olan -ve bence her mühendisin aklında olması gereken- bu temel soruyu da düşünmeye çalıştım. Dünyadan savaşlar hiçbir zaman silinmedi. Üretilen her türlü bilginin kötüye kullanilmasi da bitmedi. Bunun olduğu durumda yeniyi arayanı degil, yeniyi kötüye kullananı suçlamak gerekir. Bunca zamandır insan gerçekten de gelişememiş ise, belki de ”yeni zaten yoktur”. Yeni yoksa, yeni bir silah için bu çalışmada bilgi de üretilmemiş demektir. Bu yüzden artık korkmuyorum.

Savaş ekonomisini ve bilerek bu ölüm makinesinde dişli olmuş mühendisleri lanetliyorum. Onlardan olmayacağımı sizlerle paylaşmak istiyorum. Silah üretmeyecek, silah kullanmayacağım.

◊

It is both refreshing and thought-provoking to write the last sentences of a study which cost you much time and labor. Man lives for many thousands of years now, unable to reach a formula of happiness and prosperity, speaks of developed humanity, today in a world, in many places man killing man; however the man he is, who was not able to solve his simplest problem: to share. The question to ask for this man can be: ”Why is the urge to search for the new, if in a world as such, all new is used for evil?”

As i was preparing this thesis, i tried to think around this old question, which was in my mind for years (and i think this very question should occupy minds of all engineers). The world was never free of wars. As the evil use of every single bit of knowledge has not ended. It is not those who prepare new, but those who use the new in name of evil; those must be the ones to blame. As man in such long time of survival is not really developed, perhaps ”new is not existent”. If new does not exist, there is no knowledge produced in this study for a new weapon. Therefore am not afraid.

I condemn the war industry and all the engineers which conciously become gears of this death machine. I want to share with you that i shall not be a part of those people. I shall not produce, i shall not use any weapon.

Besonders möchte ich mich bei Herrn Professor Martin Buss wegen seiner freundliche Vorgehensweise gegenüber den Studenten bedanken. Dies hat die komplizierte und schwierige Lehre sehr unterhaltsam gemacht.

Meine Betreuerin Angelika Peer hat mit ihrem Verständnis und ihre Hilfe den Grundstein für diese Arbeit gesetzt.

Ohne meine liebe Familie, deren Unterstützung und Liebe sowie ohne meinen alten Freund Sinan wäre diese Arbeit nicht möglich gewesen. Ganz herzlichen Dank an Euch.

♢

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Canim ailemin desteği ve sevgisi olmadan, değerli dostum Sinan olmadan bunları başaramazdım. Size de candan teşekkürler ederim.

München, den 03.11.2005
Furkan Eke

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Chapter 1

Introduction

In this study, the main goal was to design a position controller for an arm-grounded dextereous telemanipulation setup. Robotic hand movements are often categorized into predefined high level tasks like cylindrical grasp, spherical grasp, pinch and the like [PS89, Jau95]. A rigid programming of predefined tasks is possible for standalone robots for routine jobs. However, in telemanipulation, the commands to the robotic manipulator are delivered online from the operator. As an alternative to the higher task definitions, use of shape primitives serves a functional mapping of the workspace [MKCA03].

Therefore, unlike a programming of some predefined higher tasks like the ones mentioned above, an online mapping of fingertip cartesian coordinates is tried. The proposed mapper is meant to fulfill the cylindrical, spherical and conical grasp tasks. This restriction on possible tasks is inherent from the kinematics of the used robotic hand which is analysed in chapter 4.

A physical restriction inspired from natural multibody systems, like those of human arm and hand, a global criterion for convexity is tried. This idea is used to overcome multiple solutions in the joint space. This criterium is inspired from of a restriction of the configuration space angles into a maximum $90°$ range.

The research area is an active one and there are many groups working on the thema. It is not possbile to list all, but Cutkosky delivers the most similar setup to our target setup. Some of his studies are [WGC00].

1.1 Simulation for Position Mapping

The setup explained in section 1.2 is only simulated in our work. The joint angle ranges, link lengths and kinematical rules of the robotic hand are used as given by the producer.

Kinetics, and hence the potential-kinetic energies are not considered in the simulation. As mass and velocity related analysis is not considered, interaction scenarios, force-feedback, inertia and stiffness are not modelled.

The aim of the simulation was to find

1. a best matching backward solution of the robotic hand and
2. a proper choice of desired locations vector from human hand model which will keep human and robotic hand workspaces as similar as possible.

For the sensory glove, there is a resolution of 8-bit per joint. Human hand model joint angle ranges are used with the joint angle resolution of the sensory glove.

1.2 Target Setup

Telemanipulation setups are splitted at the highest level into two sections, the first being the operator side and the second the teleoperator side. In a telemanipulation scheme with haptic feedback, the forces and positions are not only sent one way, i.e; from operator to the teleoperator, but a force feedback from the interaction of the teleoperator to the operator is also supplied.

Teleoperation system structure assumes explicit human intervention into the technical system. Because of distinct kinematics of the two sides of the target teleoperation setup, a perfect mirroring of the operator on the teleoperator side is not met. The differences can easily result in conflicting systems which can lead to instabilities.

These conflicts are mostly left to the operator's ability to cope with those cases. The operator should learn new skills on how to avoid instabilities. The design of telemanipulation systems becomes a task of reducing the conflict cases to an amount which can be corrected or avoided by operator intervention.

As we have only developed a simulation of the target setup, the details of the hardware and the target usage are left in appendix section A for curious reader.

Chapter 2

Problem Statement

The human hand model is a 20D kinematic model and the robotic hand is a 8D kinematic model. These models will be introduced shortly, but for the beginning, we have to build a mathematical notion as first. Here are the main vectors of the problem set:

- $\underline{q}_h \in \mathbf{C}^{20}$: Human hand joint values vector
- $\underline{w}_h \in R^{18}$: Human hand fingertip cartesian coordinates vector
- $\underline{q}_b \in \mathbf{C}^{8}$: Barrett hand joint values vector
- $\underline{w}_b \in R^{12}$: Barrett hand fingertip cartesian coordinates vector

Every person has differences in hand dimensions and usage. To tackle this variability in the system, a data collection session should be carried out, which will adapt the abstract human hand model to the operator hand. Because of different sizes and usage, the mapping function g should scale and shift the workspace of each operator $\boldsymbol{W}_h$ to approximate the robotic hand workspace $\boldsymbol{W}_b$ as close as possible.

2.1 Mathematical Notion

Problem Statement 1 *Given a 20 manifold $\underline{q}_h$ and an 8 manifold $\underline{q}_b$, find a projection - reduction methodic to sort point sets*

$$\boldsymbol{W}_h = \{\underline{w}_{h_1}, \underline{w}_{h_2}, \underline{w}_{h_3}, \underline{w}_{h_4}, \underline{w}_{h_5}, \underline{0}| \in R^3\} \quad \textit{and} \quad \boldsymbol{W}_b = \{\underline{w}_{b_1}, \underline{w}_{b_2}, \underline{w}_{b_3}, \hat{\underline{0}}\,| \in R^3\}$$

whose temporal flow of $\boldsymbol{W}_h$ and $\boldsymbol{W}_b$ are syncronised with the temporal flow of the two respective configuration space vectors $\underline{q}_h$ and $\underline{q}_b$, so that the distance $\boldsymbol{W}_h - \boldsymbol{W}_b$ is minimized across time.

In order to analyse this optimization problem, let us define first

$$\begin{array}{lll} f: & \underline{q}_h \in \mathbf{C}^{20} & \rightarrow \quad \underline{q}_b \in \mathbf{C}^8 \\ g: & \boldsymbol{W}_h \in R^{18} & \rightarrow \quad \boldsymbol{W}_b \in R^{12} \end{array}$$

f being the governing I/O function of the teleoperation system which has to synchronize the two sides and g being the coupling between $\boldsymbol{W}_h$ and $\boldsymbol{W}_b$ in cartesian space.

Before beginning the design of g, the forward kinematics of the specific setup we use can better be expressed in functions so that we can explore the temporal flows of $\boldsymbol{W}_h$ and $\boldsymbol{W}_b$. The forward kinematics functions of the two sides are the following mappings

$$\begin{array}{lll} k: & \underline{q}_h \in \mathbf{C}^{20} & \rightarrow \quad \boldsymbol{W}_h \in R^{18} \\ l: & \underline{q}_b \in \mathbf{C}^8 & \rightarrow \quad \boldsymbol{W}_b \in R^{12} \end{array}$$

Forward kinematics of the two systems are derived with help of coordinate transformation matrices relating to Denavit-Hartenberg conventions as explained in [Sch03]. The forward kinematics functions $k(\underline{q}_h) = \boldsymbol{W}_h$ and $l(\underline{q}_b) = \boldsymbol{W}_b$ describe nonlinear mappings based on known geometrical rules and therefore taken to be firmly defined. For derivation of forward kinematics functions k and l see sections 3.1 and 4.1.

Functions k and l build multi-body relativity with respect to two distinct reference points, which are the respective origins, namely $\underline{0} \in \boldsymbol{W}_h$ and $\hat{\underline{0}} \in \boldsymbol{W}_b$.
The governing function of the teleoperation system will than be

$$f = k \circ g \circ l^{-1}$$

Here l^{-1} denotes the inverse kinematics of the second system. The coupling function g has to optimize the mapping not only relating the two sets' $\underline{w}_i$'s, but also origins. Actually, the internal rules between the elements of $\boldsymbol{W}_h$ and $\boldsymbol{W}_b$ are not modifiable. This reduces the optimization problem into the adaptation of g.

There are 2 jobs g must accomplish:

- to reduce $\boldsymbol{W}_h \in R^{18}$ into a meaningful $\boldsymbol{W'}_h = \{\underline{w'}_{h_1}, \underline{w'}_{h_2}, \underline{w'}_{h_3}, \underline{0} \,| \in R^3\}$.

$$r: \quad \boldsymbol{W}_h \in R^{18} \quad \rightarrow \quad \boldsymbol{W'}_h \in R^{12}$$

- and to find an optimal origin-matcher transformation

$$m: \ \underline{0} \in R^3 \ \rightarrow \ \hat{\underline{0}} \in R^3$$

(2.1)

Here r can be differently chosen. $\boldsymbol{W'}_h$ can be built with simple weighting of elements of $\boldsymbol{W}_h$ or a geometrical set can be based on those. See section 7. Origin-matcher transformation uses 3 rotational and 3 translational parameters. Moreover, to compensate for the size mismatches, a scaling function s can be used. Than the computation sequence can look like

$$g = r \circ s \circ m$$

Functions r, s and m are neither sequential, nor parallel. They are ”in-another”. They are one. Therefore the optimization algorithm should adapt all in the same program.

Chapter 3

Human Hand Model

Based on some of the ideas in appendix section **??**, a human hand model is derived. With coordinate transformations from world frame to the links of the hand and back, the points along the finger links are found. These are expressed in cartesian coordinates and are visualized with a 3D plot. The world (ground) reference frame is chosen to have its origin at the center of the palm, with x axis pointing towards the wrist, y axis parallel to the base points of the digit fingers pointing in the thumb and the z axis pointing from the palm center upwards. The palm is assumed to be planar and to have no bending movement.

The model kinematics is seen in figure 3.1 and the corresponding 3D plot is visualized as in figure 3 for some postures. Human Hand model is based on the anatomical definitions of the joint groups [Lip00]. Barrett Hand's 20D joint space is represented with the joint vector

$$\underline{q}_h = \begin{bmatrix} q_{1_abduct} \\ q_{1_prox} \\ q_{1_media} \\ q_{1_distal} \\ \hline q_{2_abduct} \\ q_{2_prox} \\ q_{2_media} \\ q_{2_distal} \\ \hline \vdots \\ \hline q_{5_abduct} \\ q_{5_prox} \\ q_{5_media} \\ q_{5_distal} \end{bmatrix} \begin{array}{l} \}1\text{Index Finger} \\ \\ \}2\text{Middle Finger} \\ \\ \}3\text{Ring Finger}; 4\text{Little Finger} \\ \\ \}5\text{Thumb} \end{array} \tag{3.1}$$

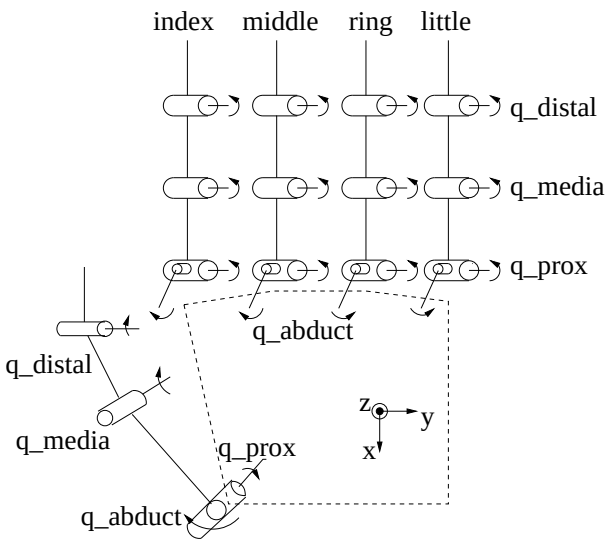


Figure 3.1: Human hand kinematics

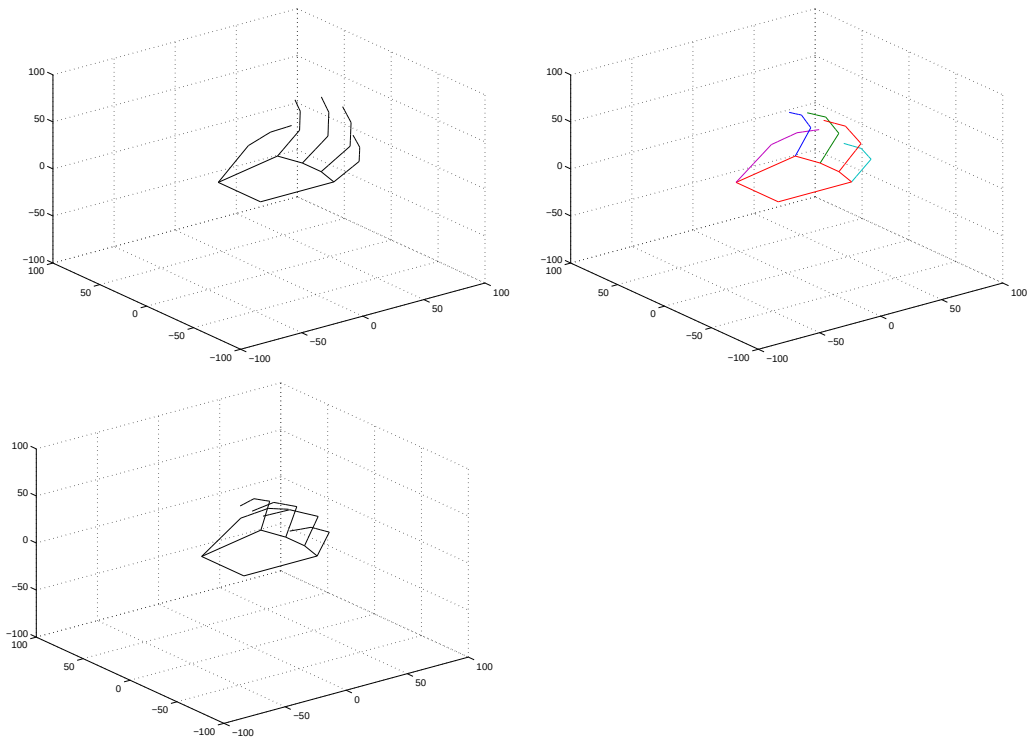


Figure 3.2: Human hand model

3.1 Forward Kinematics of Human Hand, k

The function k as already defined in section 2, corresponds to the human hand forward kinematics in our specific problem. The structure of the function k will be a chain of homogeneous transformation matrices. All transformations are applied to the current reference frame; therefore, new transformations are always left-multiplied. Here as an illustration and because of their special importance, only the forward kinematics of the index and thumb will be derived. For the complete forward kinematics calculations of human hand model, see appendix section B.
Index and thumb base reference frames are

$$
\begin{aligned}
{}^{0}T_{1_base} &= trans(1_{base_x}, 1_{base_y}, 0) \\
{}^{0}T_{5_base} &= trans(5_{base_x}, 5_{base_y}, 0)
\end{aligned}
$$

For the proximal link points, a double rotation once for flexion and once for abduction of the proximal link at the base point is used. Flexion (closing) movement is a rotation around the y axis, and the abduction movement occurs around the z axis. All the coordinate transformations are done with respect to the actual frame of reference and not with respect to the world frame. Therefore, successive coordinate transformation matrices are left multiplied.

$$
{}^{1_base}T_{1_prox} = rot(y, \tfrac{\pi}{2} + q_{h_1_prox}) rot(z, q_{h_1_abduct})
$$

The thumb requires special attention. Free position of the thumb is much different than that of the digit fingers. The angular offsets are much higher and because of the base frame built as in B.1, the axis of rotation will be different as well. Thumb base frame will be rotated with offsets, the joint movements will be added onto the offset rotation. The offsets are to be calibrated according to the free positioning of the thumb of the operator.

$$
{}^{5_base}T_{5_prox} = rot(x, q_{h_5_prox} + \tfrac{\pi}{18}) \quad rot(z, q_{h_5_abduct} - \tfrac{3\pi}{4}) \tag{3.2}
$$

The thumb can be moved both parallel and perpendicular to the palm. This is the main reason of difficulty in imagination and modeling of thumb behavior. The digit fingers do not have such a large abduction freedom and digit abduction is of secondary importance as compared to flexion. In thumb this is not the case and the both degrees of freedom of thumb base joint are of equal importance. A second difficulty in modeling thumb is the offset rotations, which makes imagination of a nonorthogonal frame shift difficult. However, the hand model will not be simplified at this stage. The simplifications will follow during the design of mapping g.

The proximal link points of the digit fingers are along the z axis of the corresponding prox frame. This choice is also simpler for the generation of the next set of frames for the media and distal link frames. Flexion movement occurs again around the y axis after a translation along the z axis of the previous frame with a length of proximal link.

$$
\begin{aligned}
{}^{1_prox}T_{1_media} &= rot(y, -q_{h_1_media}) \quad trans(0, 0, l_{1_media}) \\
{}^{5_prox}T_{5_media} &= rot(z, q_{h_5_media}) \quad trans(0, l_{5_media}, 0) \\
\\
{}^{1_media}T_{1_distal} &= rot(y, q_{h_1_distal}) \quad trans(0, 0, l_{1_distal}) \\
{}^{5_prox}T_{5_media} &= rot(z, q_{h_5_distal}) \quad trans(0, l_{5_distal}, 0)
\end{aligned}
$$

Then points along any link of any finger will be found with a translation along the translation axis. The resolution of points along the links are determined with the parameter *length_resolution*. If *length_resolution* is chosen to be 5, the link length is divided into 5 equidistant points. The intermediate points can be used in future work to model interaction. For instance the third raster point along the index finger media link in its own frame is

$$
\begin{aligned}
A &= trans(0, 0, l_{1_media}(length_index = 3)) \\
&\quad .{}^{1_prox}T_{1_media} \, .{}^{1_base}T_{1_prox} \, .{}^{0}T_{1_base}
\end{aligned}
\tag{3.3}
$$

The same point in the world coordinates is the translation part of A^{-1}. That is, the third point on the index finger media link is $(A^{-1}(1,4); A^{-1}(2,4); A^{-1}(3,4))$. Finger tips are derived similarly. Let us define the operator $pos()$ as one which extracts the translation part of a homogeneous transformation matrix, then the endpoints are found as follows[1]

[1]For ease of use, only the Finger tip frames in short form will be used. E_1 will stand for the index finger endpoint in cartesian coordinates, whereas ${}^0T_{E_1}$ will stand for the reference frame at the endpoint of the index finger.

$$
\begin{aligned}
E_1 &\hat{=} \underline{w}_{h_1} = (x_{h_1}, y_{h_1}, z_{h_1}) \\
&= pos(inv({}^{1_distal}T_{1_tip} \cdot {}^{1_media}T_{1_distal} \\
&\quad \cdot {}^{1_prox}T_{1_media} \cdot {}^{1_base}T_{1_prox} \cdot {}^{0}T_{1_base})) \\
E_5 &\hat{=} \underline{w}_{h_5} = (x_{h_5}, y_{h_5}, z_{h_5}) \\
&= pos(inv({}^{5_distal}T_{5_tip} \cdot {}^{5_media}T_{5_distal} \\
&\quad \cdot {}^{5_prox}T_{5_media} \cdot {}^{5_base}T_{5_prox} \cdot {}^{0}T_{5_base}))
\end{aligned}
$$

which constitute the human hand forward kinematics function

$$k(\underline{q}_h) = \boldsymbol{W}_h$$

3.2 Joint Angle Ranges

The joint angle measuring device has 8bit resolution per joint. To achieve similarity in the simulation data set, the joint angle ranges are divided into linearly distributed 128 values. In the real time operation the joint angles will only be replaced with measured bytes from the measurement device. The following digit finger proximal, media and distal joint angle ranges are taken from medical literature [Lip00]:

	Proximal	Media	Distal
offset	90°	0°	0°
max	90°	100°	45°

Table 3.1: Offset and maximum values for flexion joints of digit fingers

For the other joint angle ranges similar maximum rough values are chosen as

	Proximal	Media	Distal
offset	20°	0°	0°
max	30°	45°	45°

Table 3.2: Offset and maximum values for the joints of thumb

	Index Abduct	Middle Abduct	Ring Abduct	Little Abduct	Thumb Abduct
max	$20°$	$10°$	$10°$	$20°$	$45°$
offset	$20°$	$10°$	$10°$	$20°$	$-135°$

Table 3.3: Offset and maximum values for digit abduction joints

Beyond the static properties of the human hand, such as above given joint angle ranges and link lengths, the dynamic behavior of the human hand should also be considered to achieve the goal of minimal $g(\underline{w}_h, t) - \underline{w}_b(t)$ across time. For some observations and ideas on human hand dynamics see chapter 6.

Within the ranges of assumed human hand joint angles, the following scenarios are simulated:

1. **Hand Close:** In this mode, all the joint angle ranges are traced with a single common index in a for loop. This mode is like making a fist.

2. **Single Index Flexion:** Index finger flexion occurs like in **Hand Close** mode and the dependencies of the other digit fingers are also simulated.

Chapter 4

Barrett Hand Model

The model of the robotic Barret Hand is built with a similar fashion as human hand model and is dimensioned according to the data sheets of the manufacturer [Tow00]. The Barrett Hand kinematics model can be seen in figure **??** and its joint space vector is

$$\underline{q}_b = \begin{bmatrix} q_{1_abduct} \\ q_{1_media} \\ q_{1_distal} \\ \hline q_{2_abduct} \\ q_{2_media} \\ q_{2_distal} \\ \hline q_{3_media} \\ q_{3_distal} \end{bmatrix} \begin{array}{l} \}1\text{Left Spread Finger} \\ \\ \}2\text{Right Spread Finger} \\ \\ \}3\text{Nonspread Finger} \end{array} \tag{4.1}$$

The corresponding 3D model can be seen in figure 4.

4.1 Forward Kinematics of Barrett Hand, l

Finger base reference frames are:

$$\begin{aligned} {}^{\hat{0}}T_{1_base} &= trans(1_{base_x}, 1_{base_y}, 0) \\ {}^{\hat{0}}T_{2_base} &= trans(2_{base_x}, 2_{base_y}, 0) \\ {}^{\hat{0}}T_{3_base} &= trans(3_{base_x}, 0, 0) \end{aligned}$$

For the proximal link points, a double rotation once for flexion and once for abduction of the proximal link at the base point is used. Flexion (closing) movement is a rotation around the y axis, and the abduction movement occurs around the z axis.

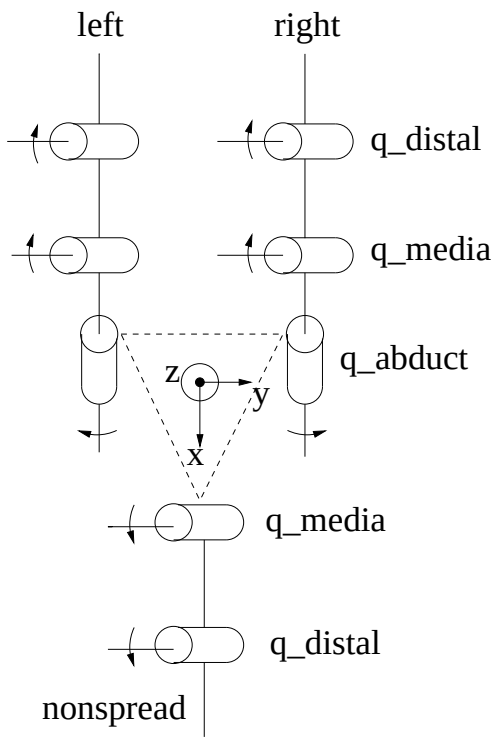


Figure 4.1: Barrett Hand Kinematics

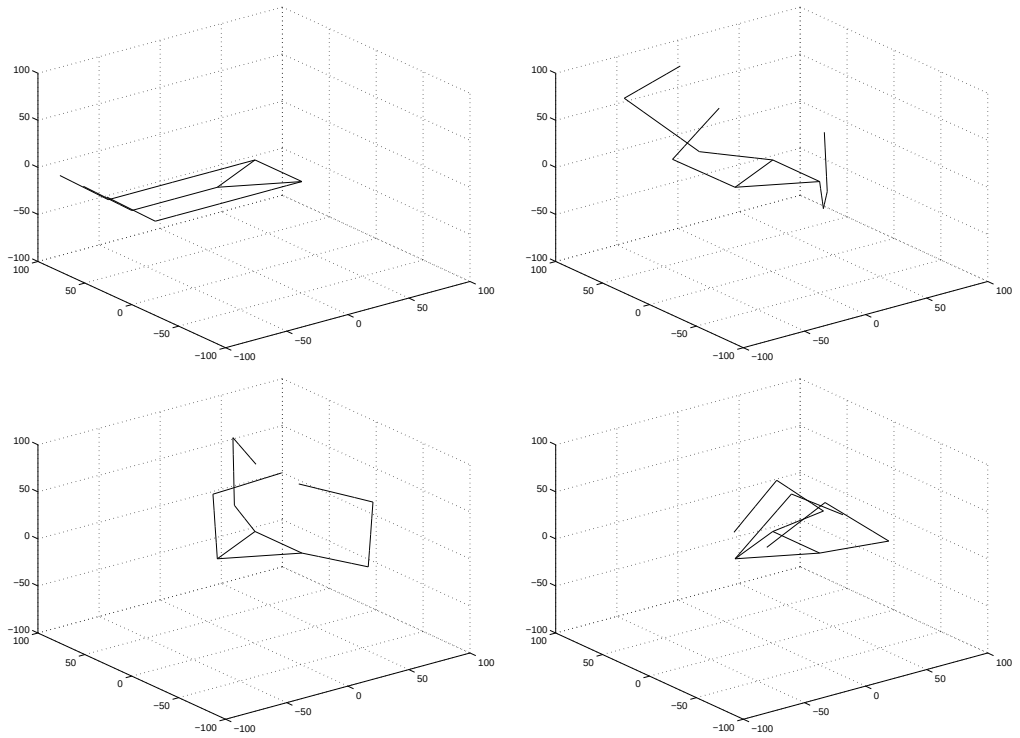


Figure 4.2: Barrett Hand model

Barrett hand digit fingers have at proximal link abduction joint only. The non-spread finger, which has no abduction ability, is taken to have no proximal link. The abduction angles of the two digit fingers are identical and are controlled with a single motor. The range of abduction angle is $0° - 180°$. But the $90° - 180°$ subrange will not be left in the mapping design.

$$
\begin{aligned}
{}^{1_base}T_{1_abduct} &= rot(z, q_{abduct} + \pi/2) \\
{}^{2_base}T_{2_abduct} &= rot(z, -q_{abduct} - \pi/2)
\end{aligned}
$$

The media links frames of the three fingers are:

$$
\begin{aligned}
{}^{1_abduct}T_{1_media} &= rot(y, -q_{1_media}) \quad trans(l_{1_abduct}, 0, 0) \\
{}^{2_abduct}T_{2_media} &= rot(y, -q_{2_media}) \quad trans(l_{2_abduct}, 0, 0) \\
{}^{3_base}T_{3_media} &= rot(y, -q_{3_media})
\end{aligned}
$$

The distal link frames of the three fingers are:

$$
\begin{aligned}
{}^{1_media}T_{1_distal} &= rot(y, -q_{1_distal}) \quad trans(l_{1_media}, 0, 0) \\
{}^{2_media}T_{2_distal} &= rot(y, -q_{2_distal}) \quad trans(l_{2_media}, 0, 0) \\
{}^{3_media}T_{3_distal} &= rot(y, -q_{3_distal}) \quad trans(l_{3_media}, 0, 0)
\end{aligned}
$$

The chained homogeneous transformation matrices from finger base points to distal joint bases are

$$
\begin{aligned}
{}^{\hat{0}}A_{1_distal} &= {}^{1_media}T_{1_distal} \;.\; {}^{1_abduct}T_{1_media} \;.\; {}^{1_base}T_{1_abduct} \;.\; {}^{\hat{0}}T_{1_base} \\
{}^{\hat{0}}A_{2_distal} &= {}^{2_media}T_{2_distal} \;.\; {}^{2_abduct}T_{2_media} \;.\; {}^{2_base}T_{2_abduct} \;.\; {}^{\hat{0}}T_{2_base} \\
{}^{\hat{0}}A_{3_distal} &= {}^{3_media}T_{3_distal} \;.\; {}^{3_base}T_{3_media} \;.\; {}^{\hat{0}}T_{3_base}
\end{aligned}
\tag{4.2}
$$

Then the finger tip points are found to be

$$
\begin{aligned}
\underline{w}_{b_1} &= pos(inv(trans(l_{1_distal}, 0, 0) \;.\; {}^{\hat{0}}A_{1_distal})) \\
\underline{w}_{b_2} &= pos(inv(trans(l_{2_distal}, 0, 0) \;.\; {}^{\hat{0}}A_{2_distal})) \\
\underline{w}_{b_3} &= pos(inv(trans(l_{3_distal}, 0, 0) \;.\; {}^{\hat{0}}A_{1_distal}))
\end{aligned}
$$

which constitute the Barrett Hand forward kinematics function

$$l(\underline{q}_b) = \boldsymbol{W}_b$$

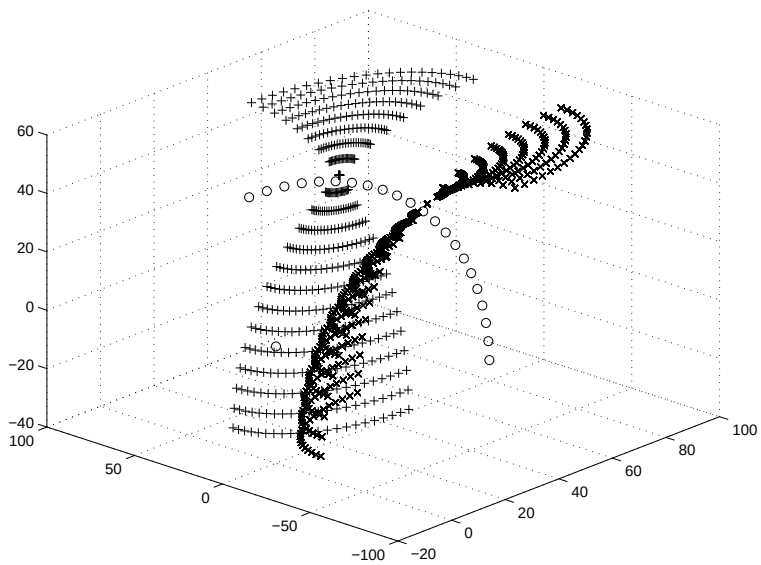


Figure 4.3: Barrett hand complete workspace, as used in the model

4.2 Internal Geometrical Rules of Barrett Hhand

The robotic hand has actually 8DOF and is controlled with 4 motors. This $8 \rightarrow 4$ mapping is achieved with mechanical gear systems and is not programmable. However, an analysis of this mechanism proves useful for the kinematic modeling of the hand.

Barrett hand finger distal joint angle limit angle is $\pi/2$. Beyond the border, the two-body mechanism behaves like a single body and the cartesian trajectory of the endpoint $\underline{w}_{b_i}$ becomes a circular arc on the flexion plane. Modifying the system equations

$$\begin{array}{lcl} f': & \underline{q}_h \in \mathbf{C}^{20} & \rightarrow \quad \underline{q'}_b \in \mathbf{C}^4 \\ g: & \boldsymbol{W}_h \in R^{12} & \rightarrow \quad \boldsymbol{W}_b \in R^{12} \end{array}$$

Controlled with only one flexion variable (one flexion motor for each finger), quasi-one-body mechanism becomes controllable.

With the symmetric spread motion, all possible movements of the two spread fingers occur on a surface in figure 4.3.

The two-body flexors are reduced into one body mechanisms. This is obviously a rough approximation to the behavior of the two link body, however the two joints are controlled with a string pulling gear drive system governed with a single motor. Therefore the 2 finger flexion links are neither separately controllable or observable.

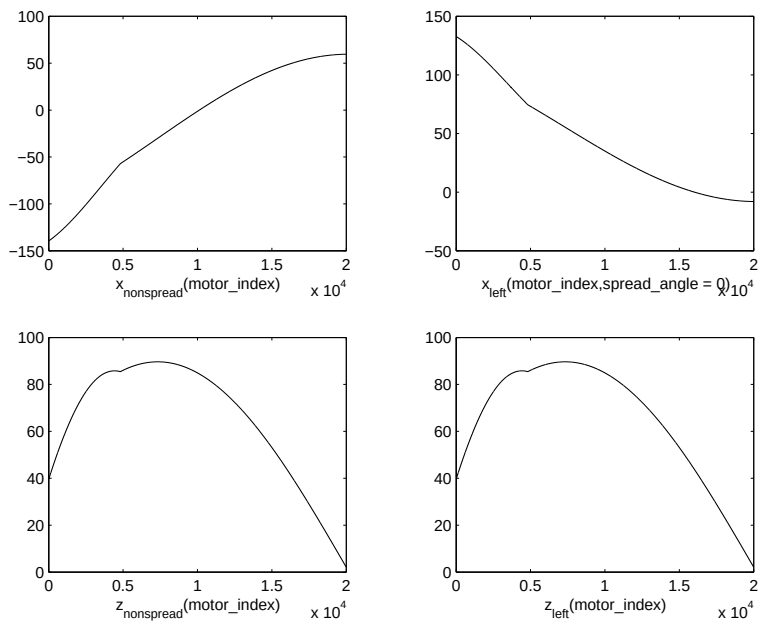


Figure 4.4: Barrett hand fingertip trajectories vs. motor encoder values

The producer nonetheless mentions that, the free space motion of the finger joints are coupled with a simple ratio [Tow00]:

$$\frac{\partial q_{media}}{\partial q_{distal}} = \frac{3}{4} \tag{4.3}$$

$$45^\circ \leq q_{media} \leq 90^\circ$$
$$0^\circ \leq q_{distal} \leq 140^\circ$$

In the simulation this fact is used together with possible joint angle ranges in derivation of possible trajectories, which can be seen in figure 4.4. If we integrate 4.3 on the distal joint angle limits

$$\begin{aligned} \int_{\pi/4}^{\pi/2} \partial q_{distal} &= \tfrac{4}{3} \int_0^{\angle COA} \partial q_{media} \\ \Rightarrow \pi/4 &= \tfrac{4}{3}(\angle COA - 0) \end{aligned}$$

Then the media joint value at which the distal joint reaches its upper limit is

$$\Rightarrow \angle COA = 3\pi/16 = q_{media}, \quad @q_{distal} = \pi/2 \tag{4.4}$$

Until $q_{media} = 3\pi/16$, 4.3 holds and the finger tip positions follow the not-circular trajectory $B \rightarrow D$. Point D is the knee point of the trajectories, which can be seen in figure 4.5. This point can be seen as a discontinuity.

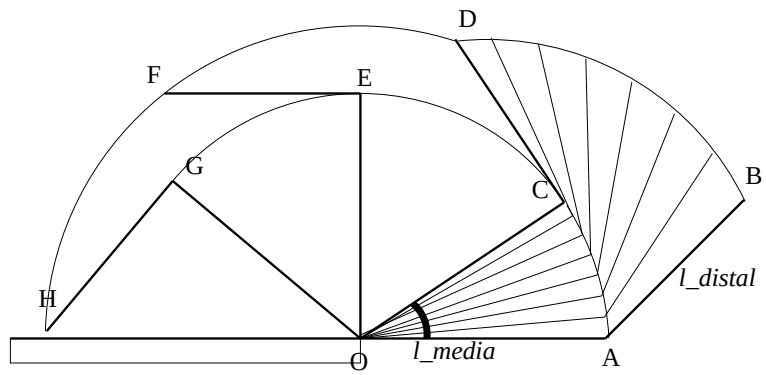


Figure 4.5: Barrett hand fingertip 2 articulation regions

Beyond the knee point D, the distal joint angle stays at $\pi/2$ and does not move any more. Than the two link body behaves like a single body following a circular trajectory. Then we can express the nonspread fingertip movement as a circular arc DFH with the radius

$$\rho = \sqrt{l_{media}^2 + l_{distal}^2} \tag{4.5}$$

Having the radius ρ of the circular workspace derived and running over all possible configurations for the spread fingers, the corresponding postures were saved. These were used in training of several networks for the numerical inverse kinematics solution.

This finger internal mapping enables a $2 \rightarrow 1$ order reduction and for the 3 fingers, a total of $6 \rightarrow 3$ order reduction. The last control variable is responsible from the spread angle and it controls the spread (abduction-adduction) angle of he digit fingers identically. This is simply a copying of the control variable onto two joints

$$\alpha_{left} = \alpha_{right}$$

This last also achieves a $2 \rightarrow 1$ order reduction and so the robotic hand-internal $8 \rightarrow 4$ order reduction is achieved.

Chapter 5

Robot Hand Inverse Kinematics

5.1 Workspace $\boldsymbol{W}_b$

The vector representing the hand finger tip points

$$\boldsymbol{W}_b = \{\underline{w}_{b_1},\ \underline{w}_{b_2},\ \underline{w}_{b_3},\ \hat{\underline{0}}\ |\in R^3\},$$

can be expressed with the notion explained in section 4.2 as follows [1]

$$\begin{aligned}
x_{b_1} &= \underline{w}_{base,1}(x) + l_{1,prox}\, cos(q_{1_abduct}) + \rho\, sin(\theta_1)\, cos(q_{1_abduct}) \\
y_{b_1} &= \underline{w}_{base,1}(y) + l_{1,prox}\, sin(q_{1_abduct}) + \rho\, sin(\theta_1)\, sin(q_{1_abduct}) \\
z_{b_1} &= \rho\, cos(\theta_1) \\
\\
x_{b_2} &= \underline{w}_{base,2}(x) + l_{2,prox}\, cos(q_{2_abduct}) - \rho\, sin(\theta_2)\, cos(q_{2_abduct}) \\
y_{b_2} &= \underline{w}_{base,2}(y) + l_{2,prox}\, sin(q_{2_abduct}) + \rho\, sin(\theta_2)\, sin(q_{2_abduct}) \\
z_{b_2} &= \rho\, cos(\theta_2) \\
\\
x_{b_3} &= \underline{w}_{base,3}(x) + \rho\, sin(\theta_3) \\
y_{b_3} &= 0 \\
z_{b_3} &= \rho\, cos(\theta_3)
\end{aligned}$$

The inverse solution l^{-1} should be able to derive the joint space vector $\underline{q}$ corresponding to the given desired end point coordinates $\boldsymbol{W}_b$.

$$l^{-1}(\boldsymbol{W}_b) = \underline{q}_b$$

The equation system which describes the endpoints set $\boldsymbol{W}_b$ for the derivation of $\underline{q}_b$ does not have separable variables and the control equations are partial differential equations. Because of this nature of the inverse analysis, a numerical inverse

[1] θ_i angles correspond to the $\angle COA$ angle in figure 4.5 and ρ is the radius from equation 4.5

solution is tried. In the conditioning of the numerical solution, the geometrical properties of the workspace will be used as explained in section 5.2

5.2 Convexness Structure of the Workspace

2 spread fingers' joint workspace $D_{spread} = D_{left} \cup D_{right}$ have 2 subspaces

$$\begin{aligned} D_{left} &= D_{left-up} \cup D_{left-down} \\ D_{right} &= D_{right-up} \cup D_{right-down} \end{aligned}$$

which can be seen as a whole in figure 4.3. So the synergic workspace of the 2 digits has the following 4 convexness structures:

$$\begin{aligned} K_1 &: \underline{w}_{b_1} \in D_{left-up} \quad \wedge \quad \underline{w}_{b_2} \in D_{right-up} \\ K_2 &: \underline{w}_{b_1} \in D_{left-up} \quad \wedge \quad \underline{w}_{b_2} \in D_{right-down} \\ K_3 &: \underline{w}_{b_1} \in D_{left-down} \quad \wedge \quad \underline{w}_{b_2} \in D_{right-up} \\ K_4 &: \underline{w}_{b_1} \in D_{left-down} \quad \wedge \quad \underline{w}_{b_2} \in D_{right-down} \end{aligned}$$

Each of the 4 convexness structures of the workspace defines a knot-free smooth manifold. A separated training of 4 subnetworks for each of the possible convexness structures can be fused into a global network of inverse solutions for 4 convexness structures. The separated training can than be achieved with separation of the I/O pairs of the inverse solution dataset into the convexness structure groups. In real time operation, the actual convexness structure can be concluded and the corresponding subnetwork can be chosen for the calculation of the inverse solution.

However, at the borders of the upper and lower subworkspaces, the convexness structure switches and the manipulability of $\boldsymbol{W}_b$ has its minimum, which can be seen as a punctual condensation of the spread finger workspaces in figure 5.2.

To avoid trajectories passing through locations of minimum endpoint manipulability, the actual convexness structure should be conserved after the first soft contact. Until reaching stable object manipulation (like grasp and hold) the actual convexness structure of the endpoint should be kept.

In our setup, the 3 fingered workspace is to be used only with the convexness structure K_1 as in figure 5.2 and therefore convexness structure switches do not occur. The numerical inverse kinematics solutions will also condition the training I/O pairs accordingly.

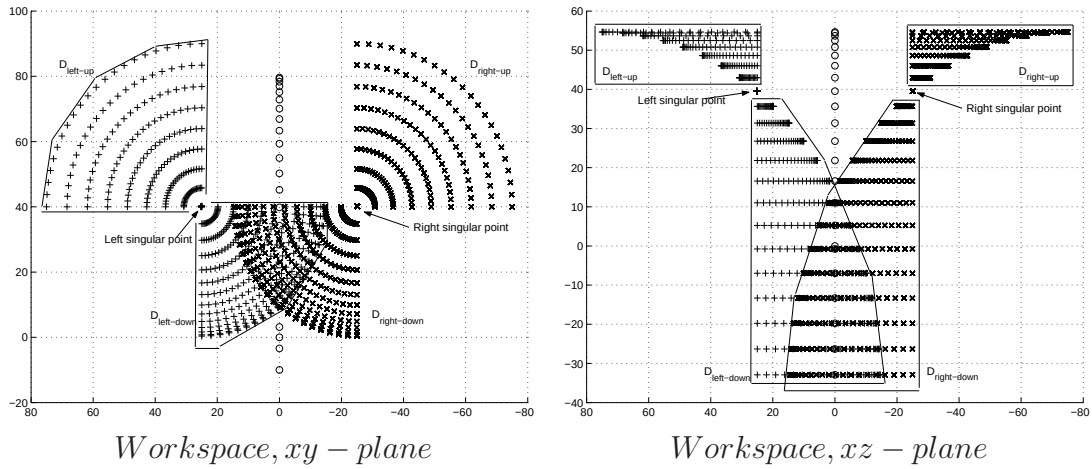


Figure 5.1: Workspace singular points

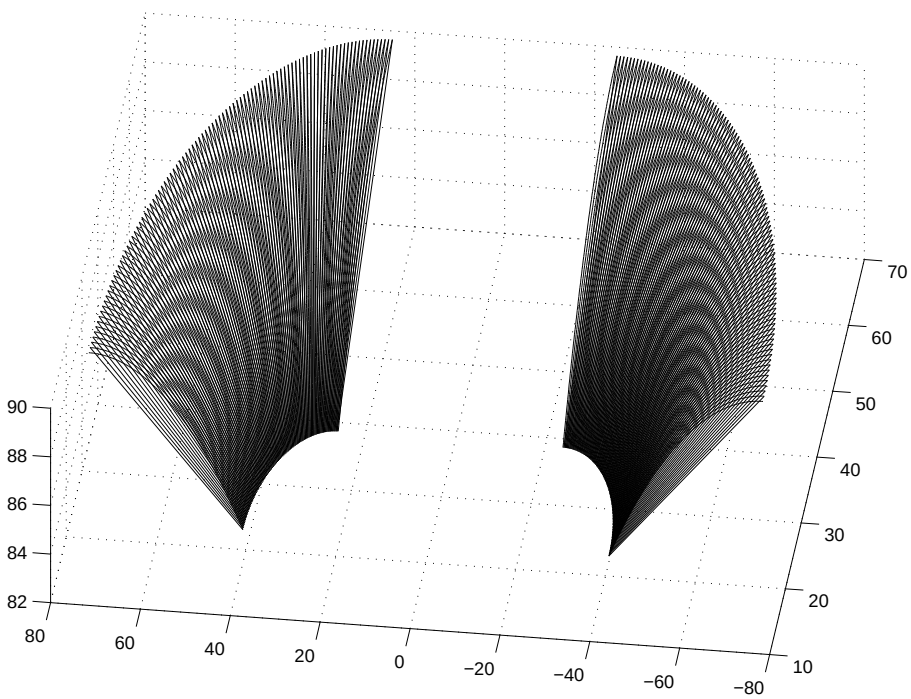


Figure 5.2: Workspace, K_1 region

5.3 Numerical Inverse Kinematics, l^{-1}

A diverse number of network structures, training algorithms and I/O pairs are used in generation of a numerical inverse kinematics function l^{-1} for bhand. Suitable template network types and learning algorithms from the Matlab Neural Network Toolbox is used. As static networks, feedforward perceptron networks with 2, 3 and 4 hidden layers are used. For dynamic solutions, in which the temporal sequence of the inputs were also important, Elman type dynamic networks are used [Elm90].
Training sets are build with forward kinematics, tracing 4D $\underline{q}_b$ configuration space. This trace can

1. start at a random posture, or a fixed posture;
   - If the initial posture $\underline{w}_b(t = 0)$ corresponding to the initial joint vector $\underline{q}_b$ is let to be random, the training should be done for several different possible initial postures. This case resembles random trajectories like open-ended strings in the workspace
   - If $\underline{w}_b(t = 0)$ is kept fixed, then the possible training trajectories of random steps constitute tree structure in the workspace.
2. Otherwise, the trace can undergo jumps of different widths in configuration space. Maximum jump width $\dot{\underline{q}}_{b_{max}}$ can be set up prior to training. The greater is the parameter, the higher is the net error. Good training results with high values of maximum jump width would mean a robust function.

In all the configurations, network error was derived with the mean square error function.

5.4 Training with Static Networks

Static networks use batch processing of the training pairs and a cumulative network error is built without considering the sequence of the input pairs. Hence in static networks, the sequence of the pairs is not important. There is no time aspect.

To see the training curves and their convergence behavior, see section C. In all of the training in table 5.1, where GD is gradient descent, RP is resilient propagation, GDX is gradient descent with adaptive learning step. As an example, Training 9 uses resilient propagation algorithm on a network with 3 hidden layers, with a 27000 pairs traning set which holds points of the complete flexion range. The Training 9 has done 950 training iterations and achieved a cumulative mean square error (MSE) of 0.00004077.

- All networks are of type feedforward perceptron networks.

Training	Algorithm	Net	Epochs	TrainSet	MSE
1	GD	12	3000	8000	0.170524
2	RP	12 12	3000	8000	0.0097950
3	GDX	16 12	3000	8000	0.0630721
4	RP	48 24	400	8000	0.0069576
5	RP	160 72 24	150	8000	0.0044750
6	RP	320 144 48	450	8000	0.0029029
7	RP	320 144 48	1500	8000	0.0009240
8	GDX	320 144 48	1500	8000	0.0006977
9	RP	144 320 48	950	27000	0.00004077
10	RP	320 144 48	1500	8000, K_1	0.0005259

Table 5.1: Training into static networks

- Training sets which span complete flexion workspace are used. Therefore all convexness structures are trained together. In Training 10 training pairs rastered the K_1 region.

Remarks on static networks:

- GDX is more unlikely to sit in local minimum.
- RP algorithm converges very fast and takes much less processing time.
- First unfolding, than fusing network structure seems to give better convergence as compared to fusing-only structure: compare 6, 7, 8 with 9.
- Because of batch processing, the order of the training pairs is not important. This is examined by shuffling the training pairs. The feedforward networks build only static mappings.
- Training sets, the increase in resolution increases the separability of the output variables and gives lower MSE.
- In all configurations, the convergence experiences a sharp descent followed with a flat descent. This flat descent corresponds to overtraining; which estimates an output variable better, and the others worse. The cumulative error measure can sink but, some of individual error rates increase. After some reasonable number of epochs in the flat convergence region, the training must be stopped to avoid overtraining. The flat region of overtraining can be observed in the logs of error measures.
- There is an enhancement in convergence if the training set is restricted to K_1 region. Compare 7 and 10.

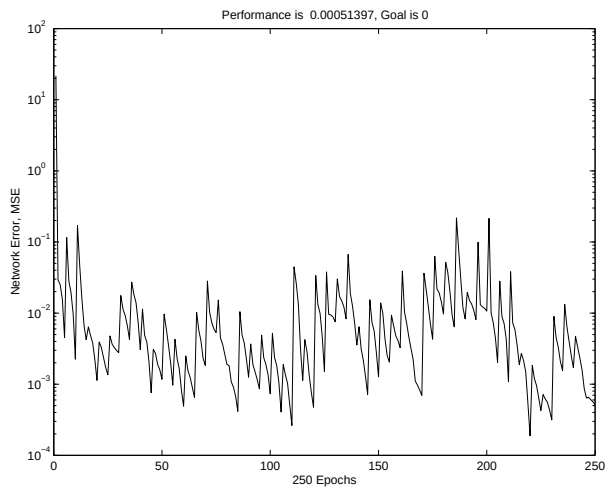


Figure 5.3: Elman dynamic network training for different trajectories

- For good results, one has to use more iterations and more training pairs and bigger networks. However, as these grow, the computation time grows drastically. Training 9, for instance, for which the best performance is achieved has locked the program. It takes roughly 1 day to compute and needs gigabytes of virtual memory.

5.5 Training with Dynamic Networks

With dynamic Elman networks, the sequence of the training pairs is also important. Discontinuities in rectangular raster training set disturb the convergence of a dynamic network training. Therefore, for the dynamic network, a training set which explores the workspace with random jumps in random directions prepared. With this set of training data, the number of convergence epochs of an Elman network with same number of layers and neurons as a feedforward network is radically decreased.

Elman networks converge much faster with better results for spacetime data. This type of network is more proper to model the inverse kinematics for the temporal flow of $\underline{w}_b$. With elman networks tree type trajectories are trained. These all have the identical initial posture and explore the workspace K_1 region randomly with maximum jump width $\underline{\dot{q}}_b \simeq 15°$. Used network has 3 hidden layers with 240,480, and 128 neurons on each. In a training session many such trajectories are trained, so the cumulative training shows periodical worsening jumps which correspond to the begin of a new trajectory in figure 5.3.

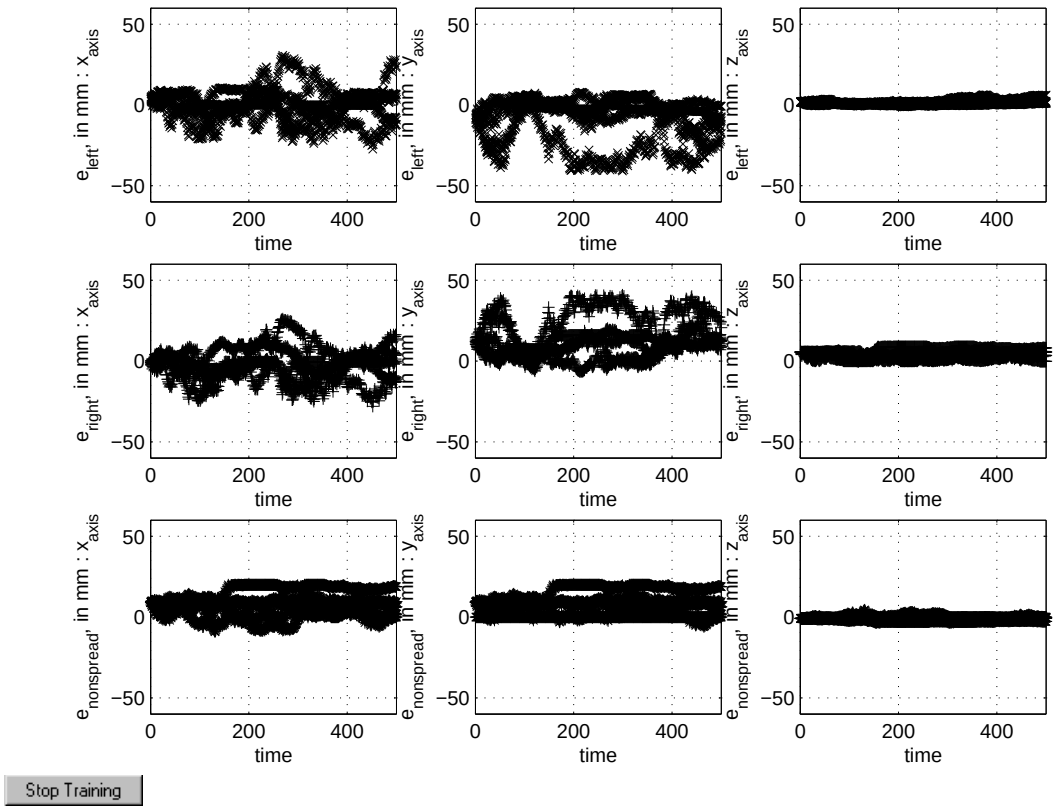


Figure 5.4: Elman dynamic network estimation errors

The training curve does give a measure of cumulative error measure built on mean square error of the network output vector. This means, one can not conclude a good evaluation of the function approximation of the network by looking simply on the training curve. For that reason, the output of the trained inverse solution is fed back to the barrett hand model to compare the errors of the individual fingertips. In figure 5.4 the representation ability of the trained kinematical structure is visible. The curves are generated for 5 random test trajectories.

5.6 Results

Static networks do not consider time, whereas dynamic networks also learn the temporal flow inside workspace. The training time is very high (1day) in static networks. The training time is much lower (few hours) in Elman networks with exploring dataset. This is mainly because, the exploring dataset has much less number of training pairs which enabled fast convergence. Dynamic networks learns temporal flow of the postures. Therefore these are more suitable to learn operator workspace behavior. So that the exploring dataset can also reflect the operator behavior.

Chapter 6

Physiological Information

6.1 Finger Movements

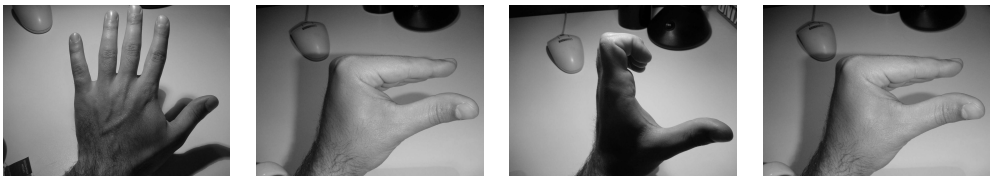
Human hand finger movements can be roughly categorized into hand-internal and into finger-internal movements. Finger internal movements can be further analyzed into scara-only (only flexion-extension) and non-scara (flexion-extension and abduction-adduction) scales. Afterwards, one can define higher level tasks a hand can accomplish.

Some of the important facts on finger movements are as follows:

1. (b) Proximal Flexion: The proximal flexion is independent of the media and distal flexion movements.

2. (c) & (d) Medial & Distal Flexion: Human phalanx distal and phalanx media joints' dynamic behavior are dependent to each other, which is very similar to the functioning of the robotic finger flexion joints. In free space it is not possible to move phalanx media without moving phalanx distal and vice versa.

3. (e) & (f) Possible 2 Finger mappings, like $\underline{w}_{h_1} + \underline{w}_{h_2} \rightarrow \underline{w}_{b_1}$ and $\underline{w}_{h_3} + \underline{w}_{h_4} \rightarrow \underline{w}_{b_2}$, are not perfectly separable. A flexion of a pair implies a slight flexion on

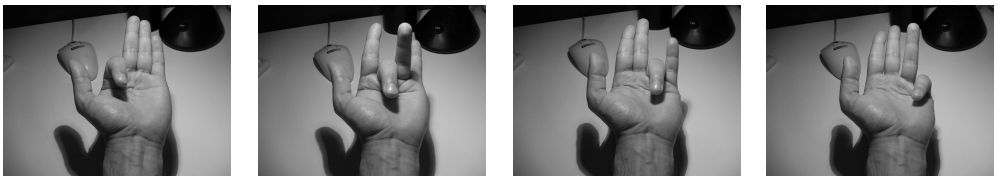


(a) null position (b) proximal (c) medial (d) distal

Figure 6.1: Synergic Flexion

(e) flexion index middle (f) flexion ring little

Figure 6.2: 2 Finger Synergic Flexion



(g) flexion index (h) flexion middle (i) flexion ring (j) flexion little

Figure 6.3: Single Finger Flexion

the immediate neighbour. Therefore digit fingers' cartesian coordinates can be averaged as grouped above.

4. (e) - (j) Induced flexion has an additive character.

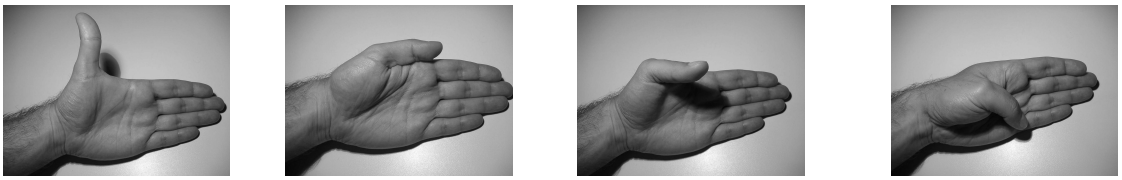
5. (e) - (j) & (l) Abduction-adduction is dependent on flexion. As the individual fingers are flexed, the neighbouring fingers undergo an induced abduction. Therefore for digit fingers abduction is of secondary importance. Full synergetic Flexion of the digit fingers reduce the abduction angle between the fingers automatically to zero at the full flexion. This fact also supports the idea that the abduction-adduction is derived from the flexion. This fact can be seen as a primitive of the human hand which can be replayed by the robotic hand.

6. (k) Extension of the fingers is not possible with the robotic hand used. Actually the robotic hand starts with a $45°$ flexion of distal joint, indeed.

7. (m) - (p) Thumb is placed very irregularly across the palm. It can move both parallel and perpendicular to the palmar plane. Thumb media and distal joints

(k) synergic extension (l) synergic full flexion

Figure 6.4: Synergic Extension and Flexion



(m) orthogonal thumb (n) parallel thumb (o) thumb distal flexion (p) thumb full flexion

Figure 6.5: Thumb Movements

are also interdependent like those of the digit fingers.

6.2 The Calibration Method

The simulation uses joint angle values which are linearly spaced inside the ranges given in section 3.2. These ranges are traced accordingly for a finger closing movement, like for instance making a fist. The joint angles used in the model are not compatible with that of the sensory glove. Main differences can be listed as the following.

- The finger distal joints are not measured.

  The distal joints are a function of the media joints. Therefore the distal joint angle can be estimated as a function of the medial joint.

  $$\underline{q}_d istal = h(\underline{q}_m edia)$$

  In order to design such a function, finger closing movements can be recorded with fast photography and the function can be approximated.

- Human hand model uses on abduction joint angle for each finger base joint. The sensory glove can measure the angles between the fingers.

  Because of this difference, an adapter software is needed. This adapter software should derive model abduction angle values out of the measured abduction angle values.

- It is possible to measure the palmar bendings, but the human hand model assumes that the palm is flat and firm.

  For the given human hand model, the palmar bending measurements are not needed. If these are not read, the bandwidth of the sensory glove can be increased. This enables better realtime operation.

After measurement of the finger link lengths from the inner side of the hand, the finger abduction start angles are to be determined. Calibration for different operators can be made with development of some gauges which will enable keeping of some predefined human hand posture $\underline{w}_{h,predefined}$. With use of such stenciling rulers and having the operator finger links measured, one can set the offsets and slopes of the multi-linear sensory device. This is not developed and an abstract human hand model is used instead.

Chapter 7

Choice of Mapping Triangle, r

We need to replicate the gestures of the human hand with the robotic hand to the highest degree possible. In order to do that, some of the difficulties are to be studied. For the three end points of the barrett hand, we have to deliver 3 points from the current human hand configuration. Difficulties in successful matching of the points from the human hand onto the workspace of barrett hand are

- the proper choice of the human hand origin,
- proper rectification of the thumb fingertip locations,

The need thumb rectification arises because, the human hand thumb is not placed on the symmetry axis of the digit fingers as is the case in barrett hand. For the fingertip trajectorial similarity to barrett hand, the following constraint can be set:

> The barrett hand nonspread finger flexion occurs in the symmetry plane of the two spread fingers. Therefore, the desired nonspread trajectory from human hand should better lay in the symmetry plane of the two other points.

Here we defined the symmetry plane of used digits and derived the projection of the thumb tip onto that plane to find nonspread desired trajectory points. This is denoted as the function p.

At the first step, the pentagon is to be reduced to a triangle with one of the following strategies:

1. **Strategy1** using the index, ring and thumb fingertips only

$$r : \{ \begin{array}{lcl} \underline{w}_{h_1} & = & \underline{w'}_{h_1} \\ \underline{w}_{h_3} & = & \underline{w'}_{h_2} \\ p(\underline{w}_{h_5}) & = & \underline{w'}_{h_3} \end{array}$$

   drawback: the palmar center is to be replaced; for instance with the triangle center

2. **Strategy2** using the index, little and thumb fingertips only

$$r : \{ \begin{array}{lcl} \underline{w}_{h_1} & = & \underline{w'}_{h_1} \\ \underline{w}_{h_4} & = & \underline{w'}_{h_2} \\ p(\underline{w}_{h_5}) & = & \underline{w'}_{h_3} \end{array}$$

   drawback: little finger is very weak and not dominant in stable postures.

3. **Strategy3** building a mean from index-middle and ring-little pairs and taking thumb tip as the third point

$$r : \{ \begin{array}{lcl} \underline{w}_{h_1} + \underline{w}_{h_2} & = & \underline{w'}_{h_1} \\ \underline{w}_{h_3} + \underline{w}_{h_4} & = & \underline{w'}_{h_2} \\ p(\underline{w}_{h_5}) & = & \underline{w'}_{h_3} \end{array}$$

   drawback: $\underline{w'}_{h_1}$ and $\underline{w'}_{h_2}$ can follow large jumps because of the independent flexion movement of the paired fingers

For the derivation of the function p see the sections 7.1 and 7.2.

7.1 Generation of a Symmetry Plane

To be able to derive a symmetry plane, we needed 3 points. Therefore, for the two points corresponding to the digit movements, 2 past tip locations were used. Together with the current tip locations we build the middle locations from left and right locations. For Strategy1,

$$\begin{array}{lcl} S(t) & = & (E_1(t) + E_3(t))/2 \\ S(t-1) & = & (E_1(t-1) + E_3(t-1))/2 \\ S(t-2) & = & (E_1(t-2) + E_3(t-2))/2 \end{array}$$

The temporal endpoint trajectory $S(t-2), S(t-1), S(t)$ constitutes an instantaneous plane of symmetry. First we need to derive a frame with vectors:

$$\begin{array}{lcl} \underline{v}_1 & = & S(t) - S(t-1) \\ \underline{v}_2 & = & S(t) - S(t-2) \end{array}$$

The normal to the plane is the cross-product of the two vectors:

$$\underline{n} = \underline{v}_2 \times \underline{v}_1$$

The projection operator of the plane with the normal vector $\underline{n} = (n_x, n_y, n_z)$ is:

$$P = \frac{1}{n_x + n_y + n_z} \begin{bmatrix} n_y^2 + n_z^2 & -n_x n_y & -n_x n_z \\ -n_x n_y & n_x^2 + n_z^2 & -n_y n_z \\ -n_x n_z & -n_y n_z & n_x^2 + n_z^2 \end{bmatrix}$$

Then the projection of the thumb endpoint on the symmetry axis is the product:

$$E_5' = P \cdot E_5$$

The orientation is also important. Therefore, instead of analytical derivation of the symmetry plane and a projection operator, a frame representing the symmetry would be more proper. With this approach the symmetry frame instead of symmetry plane is generated, and the computational complexity is remarkably reduced. As this procedure will work on online data, computational efficiency is very important.

7.2 Reference Frame Fusion

Instead of two cross products and a projection operation, a symmetry reference frame instantaneous symmetry point $S(t)$ relative to the chosen finger tip frames of the chosen strategy, can be easily derived as follows:

After the calculation of the forward kinematics of the human hand, we have all the finger tip locations in cartesian coordinates. We calculate the point $S(t)$ as before. In this procedure, we do not need the memory either: we will only use the current points.

The coordinate transformation matrices of the individual fingers left multiplied with the point S in world coordinates gives the location of the point S in finger tip frames of references.

7.2.1 Strategy3 with Weighting

Strategy3 of those described in section 7 constitutes the general case of all 3, if the individual frames and corresponding fingertip locations are fused with weighting. In this section, the Strategy3 will be derived first and later it will be shown in its weighted form.

The thumb location will be projected on the vertical symmetry plane of this derived reference frame. This way, the projected thumb trajectory will be compatible with the barrett hand trajectory, which is also inside the vertical symmetry plane of the barrett hand digit fingers.
First, find the location of the symmertry point S of the to chosen fingers' tips

$$\begin{aligned} S_{@E_1} &=^0 T_{E_1} S(t)_{@world} \\ S_{@E_2} &=^0 T_{E_1} S(t)_{@world} \\ S_{@E_3} &=^0 T_{E_1} S(t)_{@world} \\ S_{@E_4} &=^0 T_{E_4} S(t)_{@world} \end{aligned} \tag{7.1}$$

The symmetry frame will have its origin at the symmetry point S. This symmetry plane will

$$\begin{array}{rcl} {}^{E_1}T_S & = & trans(S_{@E_1}) \cdot^0 T_{E_1} \\ {}^{E_2}T_S & = & trans(S_{@E_1}) \cdot^0 T_{E_2} \\ {}^{E_3}T_S & = & trans(S_{@E_1}) \cdot^0 T_{E_3} \\ {}^{E_4}T_S & = & trans(S_{@E_4}) \cdot^0 T_{E_4} \end{array} \tag{7.2}$$

Therefore, the mean of the unit vectors of the two translated frames from the chosen fingertip frames will be averaged to find a symmetrical frame.

$${}^0T_S = \frac{{}^{E_1}T_S \cdot {}^0T_{E_1} + {}^{E_2}T_S \cdot {}^0T_{E_2} + {}^{E_3}T_S \cdot {}^0T_{E_3} + {}^{E_4}T_S \cdot {}^0T_{E_1}}{4} \tag{7.3}$$

Now we need to take the projection of the thumb fingertip E_5 onto the symmetry plane of the frame S. The symmetry plane of the frame at S is the x-z plane. This is the plane on which the average flexion moves. Therefore, the projection of the thumb on this plane rectifies its nonorthogonal configuration.

$$p : \{ \begin{array}{rcl} E_{5@S} & = & {}^0T_S E_{5@world} \\ E_{5@S,rectified} & = & (E_{5@S}(x), 0, E_{5@S}(z)) \end{array} \tag{7.4}$$

If we express all the points of the triangle in the S frame of reference, we get

$$\begin{array}{rcl} A_{@S} & = & \frac{E_{1@S}+E_{2@S}}{2} \\ B_{@S} & = & \frac{E_{3@S}+E_{4@S}}{2} \\ C_{@S} & = & E_{5@S,rectified} \end{array} \tag{7.5}$$

The corners of the triangle $A_{@S}\widehat{B_{@S}}C_{@S}$ constitute the desired cartesian positions for the robotic fingers. Here are some important points to be studied:

- The height of the triangle may not be delivered to the control unit, if the points A,B, and C will be given in S frame.
- Not all the points in cartesian space are reachable; only some on a circular not convex arc.

Averaged coordinates/frames can be weighted and adapted to meet the common goal of functions r and m. For instance, with weighting, frame fusion in 7.3 becomes

$$ {}^{0}T_S = \frac{{}^{E_1}T_S \cdot {}^{0}T_{E_1} \cdot v_1 + {}^{E_2}T_S \cdot {}^{0}T_{E_2} \cdot v_2 + {}^{E_3}T_S \cdot {}^{0}T_{E_3} \cdot v_3 + {}^{E_4}T_S \cdot {}^{0}T_{E_1} \cdot v_4}{v_1 + v_2 + v_3 + v_4} \tag{7.6} $$

and similarly the mapping triangle in 7.5 becomes

$$ \begin{aligned} A_{@S} &= \frac{E_{1@S} \cdot v_1 + E_{2@S} \cdot v_2}{v_1 + v_2} \\ B_{@S} &= \frac{E_{3@S} \cdot v_3 + E_{4@S} \cdot v_4}{v_3 + v_4} \\ C_{@S} &= E_{5@S,rectified} \end{aligned} \tag{7.7} $$

As an alternative the triangle $A_{@0}\widehat{B_{@0}}C_{@0}$ can be expressed in world frame. This is actually more proper to best fit the modified-weighted human hand workspace and barrett hand workspace. Therefore by transforming 7.7 back to barrett hand world frame, we get the workspace reduction function r as stated in chapter 2

$$ r: \quad \boldsymbol{W}_h \quad \rightarrow \quad \boldsymbol{W'}_h : [A_{@0} \quad B_{@0} \quad C_{@0}] $$

By using $A_{@0}\widehat{B_{@0}}C_{@0}$ instead of $A_{@S}\widehat{B_{@S}}C_{@S}$, we don't need additional transformation of the barrett hand endpoints. This means, the mapping function g takes $[A_{@0} \quad B_{@0} \quad C_{@0}]$ as input vector

$$ g([A_{@0} \quad B_{@0} \quad C_{@0}]) = \underline{w}_{b,desired} $$

7.3 Complete System Function

Given k, l, and l^{-1}, one should derive the error function e to optimize the parameters governing r, m and s. This optimization will conducted over the software model of the teleoperation system, and for adaptation for different operators, should be repeated on real setup.

For a given human hand joint vector $\underline{q}_h$ we get the desired posture for the barrett hand as follows

$$ \begin{aligned} \underline{w}_h &= k(\underline{q}_h) \\ \underline{w}_{b,desired} &= s(r(\underline{w}_h)) \\ \underline{q}_{b,control} &= l^{-1}(\underline{w}_{b,desired}) \\ \underline{w}_{b,actual} &= m(l(\underline{q}_{b,control})) \end{aligned} $$

The resulting control error function is

$$\begin{aligned} e &= \underline{w}_{b,desired} - \underline{w}_{b,actual} \\ &= s(r(\underline{w}_h)) - m(l(\underline{q}_{b,control})) \end{aligned}$$

This error function is that of HW/SW training algorithm. The algorithm can be based on gradient descent with adaptive learning step μ or simply mutual selection. Here prior to training, some initial values can be set for the parameters to optimize.

The parameters to adapt for the 3 functions which constitute the mapping function g are

- For reduction function r, one has to adapt the weighting parameters v_1, v_2, v_3, v_4, of equations 7.6 and 7.7,

- For origin-matcher function m, one should optimize $n_{roll}, n_{pitch}, n_{yaw}$ angles and x_{shift}, y_{shift}, z_{shift} of a general coordinate transformation matrix ${}^0T_{\hat{0}}$,

- For the workspace scaling function s, a scalar value ω which should multiply an identity matrix only in the rotation part

$$\Omega = \begin{bmatrix} \omega & 0 & 0 & 0 \\ 0 & \omega & 0 & 0 \\ 0 & 0 & \omega & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

  This matrix should be left multiplied with the origin-matcher transformation matrix.

$$M = \Omega\, {}^0T_{\hat{0}}$$

  To be able to transform adapt all the movements and dimensions of barrett hand with the same amount, this scaler-shifter transformation will be used as the new origin of the barrett hand. The base frames will be modified accordingly.

Hence the complete vector of adaptation is

$$\tilde{n} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ \hline n_{roll} \\ n_{pitch} \\ n_{yaw} \\ x_{shift} \\ y_{shift} \\ z_{shift} \\ \hline \omega \end{bmatrix} \begin{array}{l} \}\text{parameters of } r \\ \\ \\ \\ \\ \\ \}\text{parameters of } m \\ \\ \\ \\ \\ \}\text{parameters of } s \end{array}$$

7.4 Results and Future Work

For the adaptation of this vector a simple single genom mutation-selection algorithm with guess initial values was used. The results are not satisfactory yet. The shift does not get nonspread tip $\underline{w}_{b_3}$ under control. This is an important problem.

The spread is more dominantly controlled as compared to flexion. This latter fact is predictable from the fact that, the K_1 region of $\boldsymbol{W}_b$ has a much narrower z axis range of values as compared to xy range of values. This result is also in accordance with the error measurements of the Elman network from figure 5.4.

The foreseen structures for the functions r, s and m should be structured more elegantly to reduce the difference e over temporal flow of the barrett hand and human hand trajectories.

To achieve nonspread control, the training of 3 robot fingers can be seperated, and fused together with shape primitives instead of cartesian coordinates mapping. Then it can also be possible to scale and shift the digits' and thumb's workspaces seperately.

Chapter 8

Summary

As described in the problem statement for the derivation of the system function f first the forward kinematics functions k and l of the human hand robotic hand are derived. By choosing a constrainted smooth workspace of the robotic hand with help of the generated forward kinematics, a numerical inverse kinematics function is trained into some different type of networks. The simulations show that the results for the trained inverse kinematics function l^{-1} are satisfactory.
Than, as stated in the problem statement, the mapper function for the optimization problem is parametrised and adapted. This latter goal is not achieved with satisfaction as explained in 7.4. The mapper optimization algorithm can be redesigned and improved to achieve the control of the robotic nonspread finger.

Appendix A

Target Setup Description

The subsystems of the global system are the following:

- Human operator: Evaluates the current situation by means of the available feedback channels and decides what the next action would be. The available feedback channels are haptics and direct visual channel in the setup used.
- CyberGrasp - CyberGlove module: Constitutes the haptic interaction of the human operator with the telemanipulation system. CyberGlove presents the host computer with the joint space knowledge from the operator and the CyberGrasp displays the haptic feedback data from the host computer to the operator. This couple consists of a joint space measuring glove which can measure 22 hand articulations and a force feedback device which apply forces evenly onto each 3 joints of the fingers as a virtual display of the telemanipulation forces.
- Host Computer: Constitutes the main controller to be designed in this study, which has to bind the operator and teleoperator knowledge together into a meaningful whole. Its main duty is to define a meaningful mapping from the operator joint space measurements to the robotic hand control variables. It also transmits the force feedback knowledge from the robotic hand to the operator.
- Barrett Hand: Replicates the movements of the human hand based on the control data from the host computer. Robotic hand also returns forces of interaction back to the host computer.
- Object - Environment: defines the physical context in which the telemanipulation takes place.

This structure can be seen in figureA.1.

For the host computer, the interfaces to Barrett Hand and CyberGrasp-CyberGlove couple is available. The controller can not drive for the other cascaded modules.

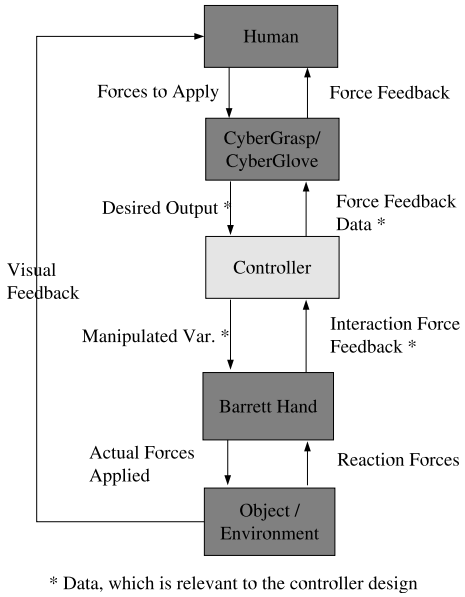


Figure A.1: System Modules

Data exchange between the host computer and the Barrett Hand is implemented in form of batch processing without a FIFO. The host computer should be able to deal with the unsuccessful control returns. The data exchange in real time operation will be implemented with 3 control and 6 feedback variables.

Data exchange between the host computer and the CyberGrasp-CyberGlove couple is implemented over the TCP/IP network protocol. Each pulling motor has a maximum force of 12N with 12bit resolution. By selecting a subset of the multi-sensor system, the maximal communication rate of the device can be increased.

Operator's finger positions are measured with the CyberGlove. This is a commercial product which has 3 flexion-extension sensors for digit fingers, 2 for thumb, 4 abduction-adduction sensors each between two neighbour fingers and two additional sensors to measure the palm bending. This sensor system assumes a multilinear configuration and has two parameters for each sensor. The first parameter is the hardware offset and the second is the gain. All the sensors used should be calibrated according to the user's hand dimensions.

Telemanipulation setup at hand is based on the collaboration of complete, input-output capable and state-space isolated modules. CyberGrasp and Barrett Hand modules are such state-isolated modules. The challenge in design of a control scheme for this system is that the inputs of the modules are nonlinearly mapped to the environment. A detailed system description is depicted in figure A.2.

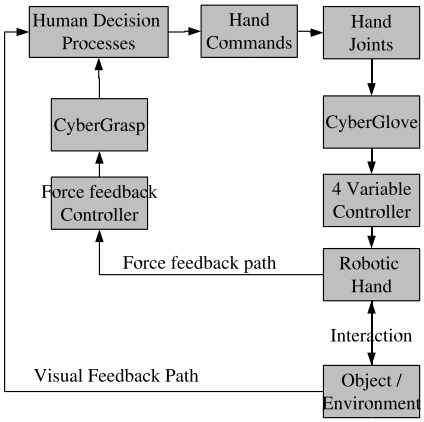


Figure A.2: System Structure

Appendix B

Human Hand Forward Kinematics Derivation

Finger base reference frames are:

$$
\begin{aligned}
{}^{0}T_{1_base} &= trans(1_{base_x}, 1_{base_y}, 0) \\
{}^{0}T_{2_base} &= trans(2_{base_x}, 2_{base_y}, 0) \\
{}^{0}T_{3_base} &= trans(3_{base_x}, 3_{base_y}, 0) \\
{}^{0}T_{4_base} &= trans(4_{base_x}, 4_{base_y}, 0) \\
{}^{0}T_{5_base} &= trans(5_{base_x}, 5_{base_y}, 0)
\end{aligned}
$$

For the proximal link points, a double rotation once for flexion and once for abduction of the proximal link at the base point is used. Flexion (closing) movement is a rotation around the y axis, and the abduction movement occurs around the z axis. All transformations are applied to the current reference frame; therefore, new transformations are always left-multiplied.

$$
\begin{aligned}
{}^{1_base}T_{1_prox} &= rot(y, q_{h_1_prox})rot(z, q_{h_1_abduct}) \\
{}^{2_base}T_{2_prox} &= rot(y, q_{h_2_prox})rot(z, q_{h_2_abduct}) \\
{}^{3_base}T_{3_prox} &= rot(y, q_{h_3_prox})rot(z, q_{h_3_abduct}) \\
{}^{4_base}T_{4_prox} &= rot(y, q_{h_4_prox})rot(z, q_{h_4_abduct})
\end{aligned}
$$

The thumb requires special attention. Free position of the thumb is much different than that of the digit fingers. The angular offsets are much higher and because of the base frame built as in B.1, the axis of rotation will be different as well. Thumb base frame will be rotated with an offset, the joint movements will be added onto the offset rotation. The offsets are to be calibrated according to the free positioning of the thumb of the operator.

$$^{5_base}T_{5_prox} = rot(x, q_{h_5_prox} + \pi/18) \quad rot(z, q_{h_5_abduct} - 3\pi/4) \tag{B.1}$$

The thumb can be moved both parallel and perpendicular to the palm. This is the main reason of difficulty in imagination and modeling of thumb behavior. The digit fingers do not have such a large abduction freedom and digit abduction is of secondary importance as compared to flexion. In thumb this is not the case and the both degrees of freedom of thumb base joint are of equal importance. A second difficulty in modeling thumb is the offset rotations, which makes imagination of a nonorthogonal frame shift difficult. However, the hand model will not be simplified at this stage. The simplifications will follow during the design of mapping g.

The proximal link points of the digit fingers are along the z axis of the corresponding prox frame. This choice is also simpler for the generation of the next set of frames for the media and distal link frames. Flexion movement occurs again around the y axis after a translation along the z axis of the previous frame with a length of proximal link.

$$
\begin{aligned}
{}^{1_prox}T_{1_media} &= rot(y, -q_{h_1_abduct}) \quad trans(0, 0, l_{1_media}) \\
{}^{2_prox}T_{2_media} &= rot(y, -q_{h_2_abduct}) \quad trans(0, 0, l_{2_media}) \\
{}^{3_prox}T_{3_media} &= rot(y, q_{h_3_abduct}) \quad trans(0, 0, l_{3_media}) \\
{}^{4_prox}T_{4_media} &= rot(y, q_{h_4_abduct}) \quad trans(0, 0, l_{4_media}) \\
\\
{}^{5_prox}T_{5_media} &= rot(z, q_{h_5_abduct}) \quad trans(0, l_{5_media}, 0) \\
\\
{}^{1_media}T_{1_distal} &= rot(y, q_{h_1_abduct}) \quad trans(0, 0, l_{1_distal}) \\
{}^{2_media}T_{2_distal} &= rot(y, q_{h_2_abduct}) \quad trans(0, 0, l_{2_distal}) \\
{}^{3_media}T_{3_distal} &= rot(y, q_{h_3_abduct}) \quad trans(0, 0, l_{3_distal}) \\
{}^{4_media}T_{4_distal} &= rot(y, q_{h_4_abduct}) \quad trans(0, 0, l_{4_distal}) \\
\\
{}^{5_prox}T_{5_media} &= rot(z, q_{h_5_abduct}) \quad trans(0, l_{5_media}, 0)
\end{aligned}
$$

Then points along any link of any finger will be found with a translation along the translation axis. The resolution of points along the links are determined with the parameter *length_resolution*. If *length_resolution* is chosen to be 5, the link length is divided into 5 equidistant points. For instance the third raster point along the index finger media link in its own frame is:

$$
\begin{aligned}
A &= trans(0, 0, l_{1_media}(length_index = 3)) \\
&\quad .{}^{1_prox}T_{1_media} \, .{}^{1_base}T_{1_prox} \, .{}^{0}T_{1_base}
\end{aligned} \tag{B.2}
$$

The same point in the world coordinates is the translation part of A^{-1}. That is, the third point on the index finger media link is $(A^{-1}(1,4); A^{-1}(2,4); A^{-1}(3,4))$. Finger tips are derived similarly. Let us define the operator $pos()$ as one which extracts the translation part of a homogeneous transformation matrix, then the endpoints are found as follows:

$$
\begin{aligned}
E_1 &\hat{=} \underline{w}_{h_1} = (x_{h_1}, y_{h_1}, z_{h_1}) \\
&= pos(inv({}^{1_distal}T_{1_tip} \cdot {}^{1_media}T_{1_distal} \\
&\quad \cdot {}^{1_prox}T_{1_media} \cdot {}^{1_base}T_{1_prox} \cdot {}^{0}T_{1_base})) \\
E_2 &\hat{=} \underline{w}_{h_2} = (x_{h_2}, y_{h_2}, z_{h_2}) \\
&= pos(inv({}^{2_distal}T_{2_tip} \cdot {}^{2_media}T_{2_distal} \\
&\quad \cdot {}^{2_prox}T_{2_media} \cdot {}^{2_base}T_{2_prox} \cdot {}^{0}T_{2_base})) \\
E_3 &\hat{=} \underline{w}_{h_3} = (x_{h_3}, y_{h_3}, z_{h_3}) \\
&= pos(inv({}^{3_distal}T_{3_tip} \cdot {}^{3_media}T_{3_distal} \\
&\quad \cdot {}^{3_prox}T_{3_media} \cdot {}^{3_base}T_{3_prox} \cdot {}^{0}T_{3_base})) \\
E_4 &\hat{=} \underline{w}_{h_4} = (x_{h_4}, y_{h_4}, z_{h_4}) \\
&= pos(inv({}^{4_distal}T_{4_tip} \cdot {}^{4_media}T_{4_distal} \\
&\quad \cdot {}^{4_prox}T_{4_media} \cdot {}^{4_base}T_{4_prox} \cdot {}^{0}T_{4_base})) \\
E_5 &\hat{=} \underline{w}_{h_5} = (x_{h_5}, y_{h_5}, z_{h_5}) \\
&= pos(inv({}^{5_distal}T_{5_tip} \cdot {}^{5_media}T_{5_distal} \\
&\quad \cdot {}^{5_prox}T_{5_media} \cdot {}^{5_base}T_{5_prox} \cdot {}^{0}T_{5_base}))
\end{aligned}
$$

[1]

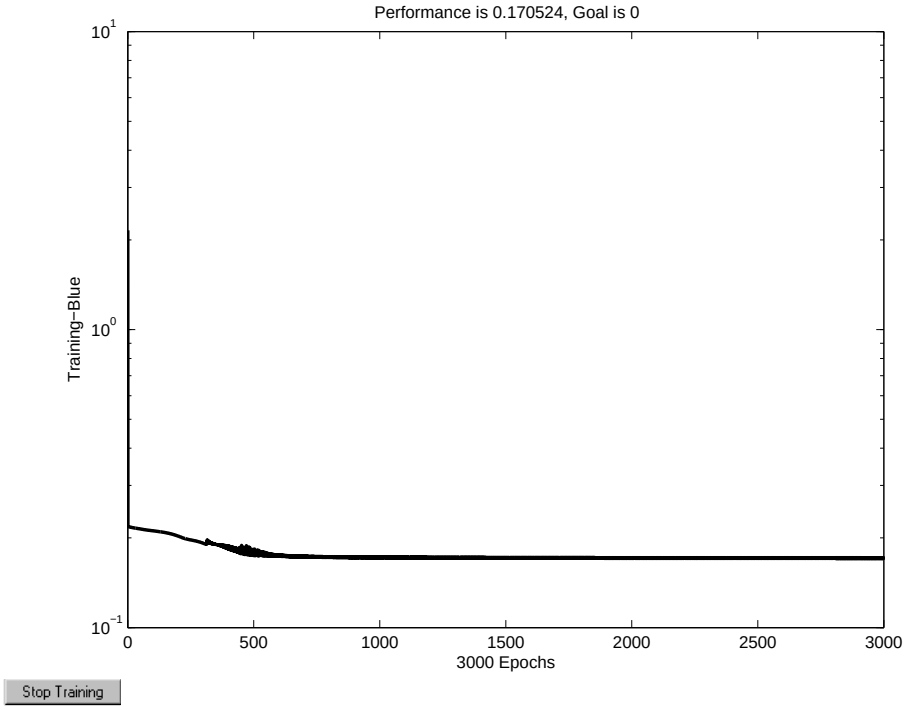
Although the immediate points before the finger tip are not used in the mapping, these points are used in visualization of the current configuration of the human hand, so that the desired postures can be validated visually. After successful design of a mapping software, calculation of intermediate points can be left out in a realtime code.

[1]For ease of use, only the Finger tip frames in short form will be used. E_1 will stand for the index finger endpoint ion cartesian coordinates, whereas ${}^{0}T_{E_1}$ will stand for the reference frame at the endpoint of the index finger.

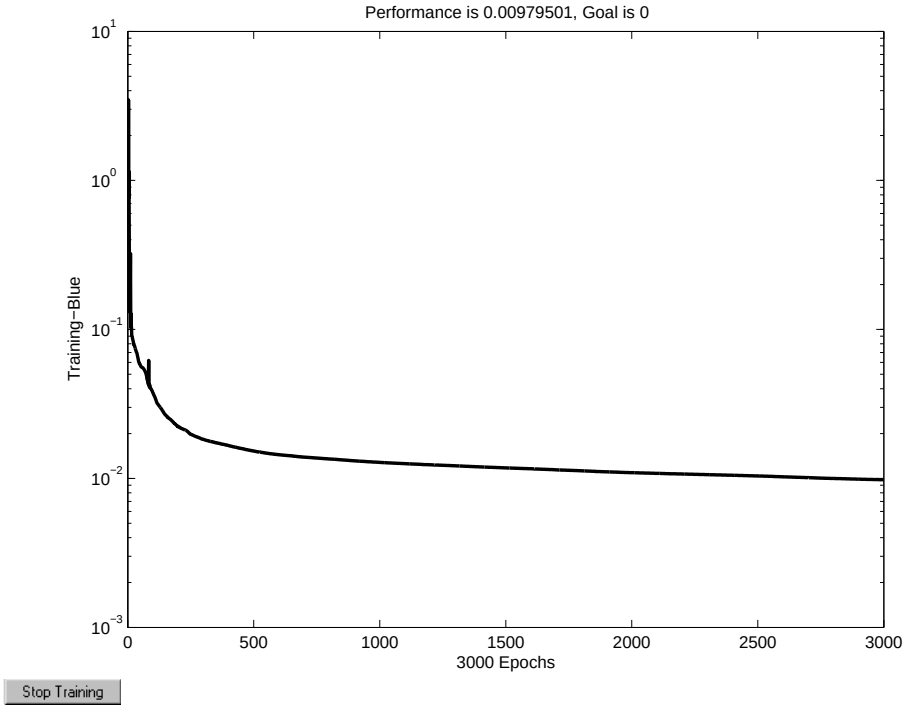
Appendix C

Training Curves for l^{-1}

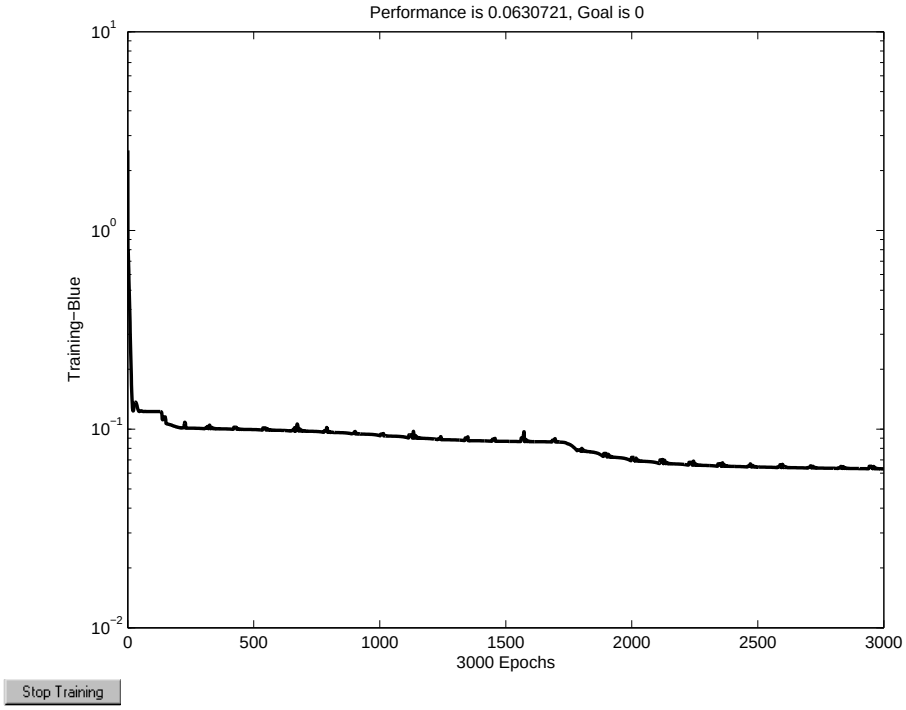
The training curves for the configurations discussed in section 5.3 are to be seen in figure C. Notice that, the last training had too much computation load and locked the program. This is the rescued image of the training curve.



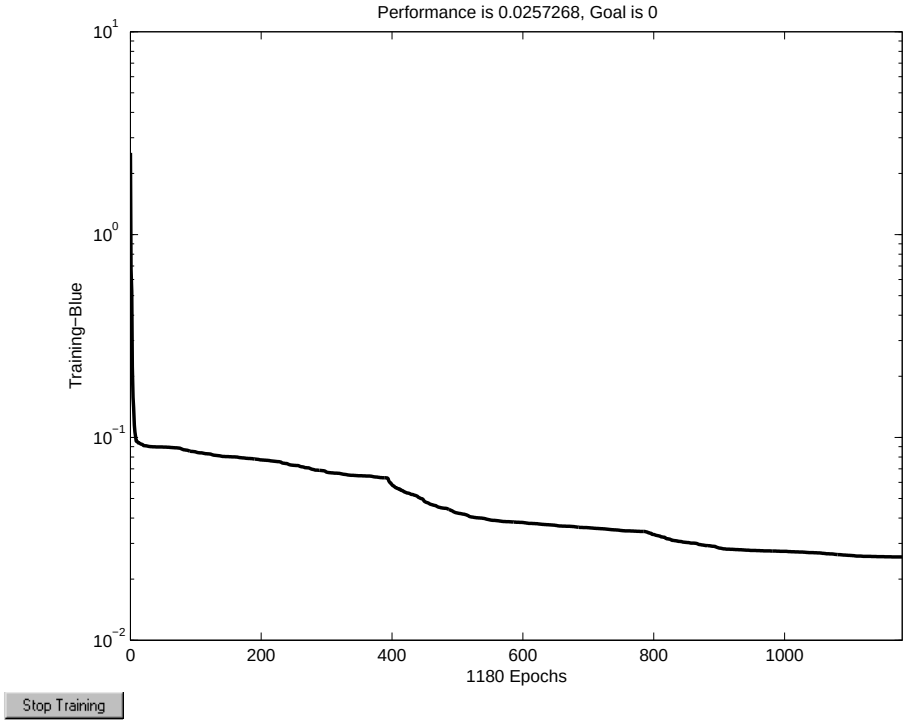
Training 1



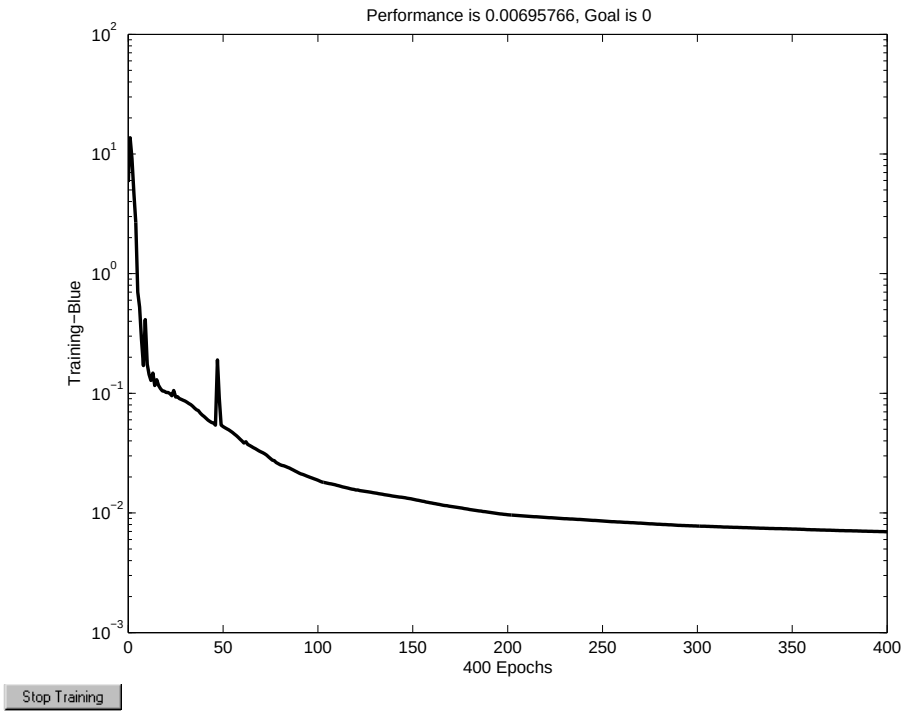
Training 2



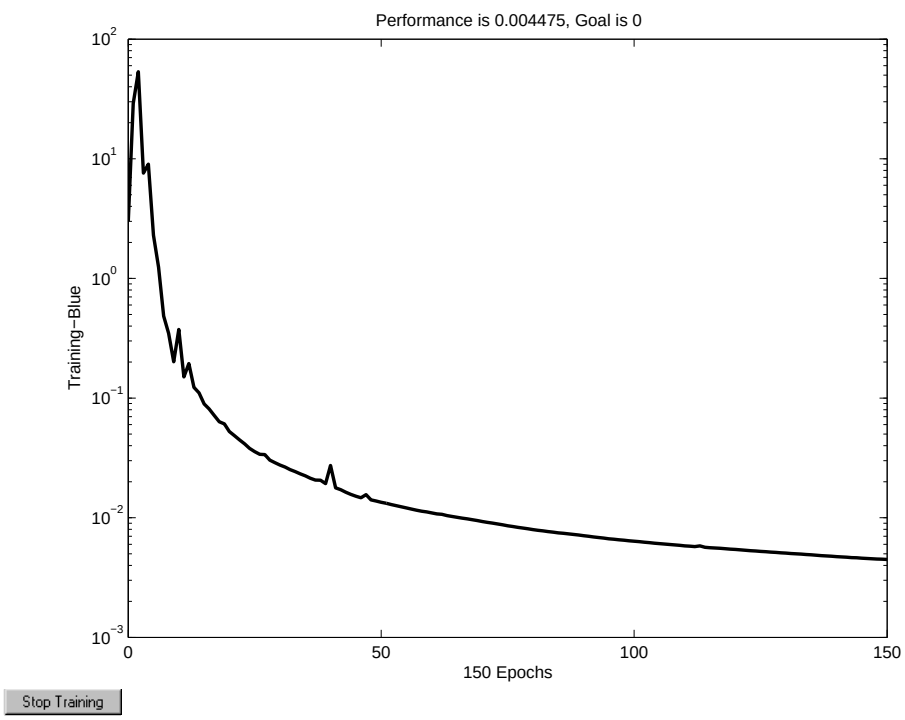
Training 3



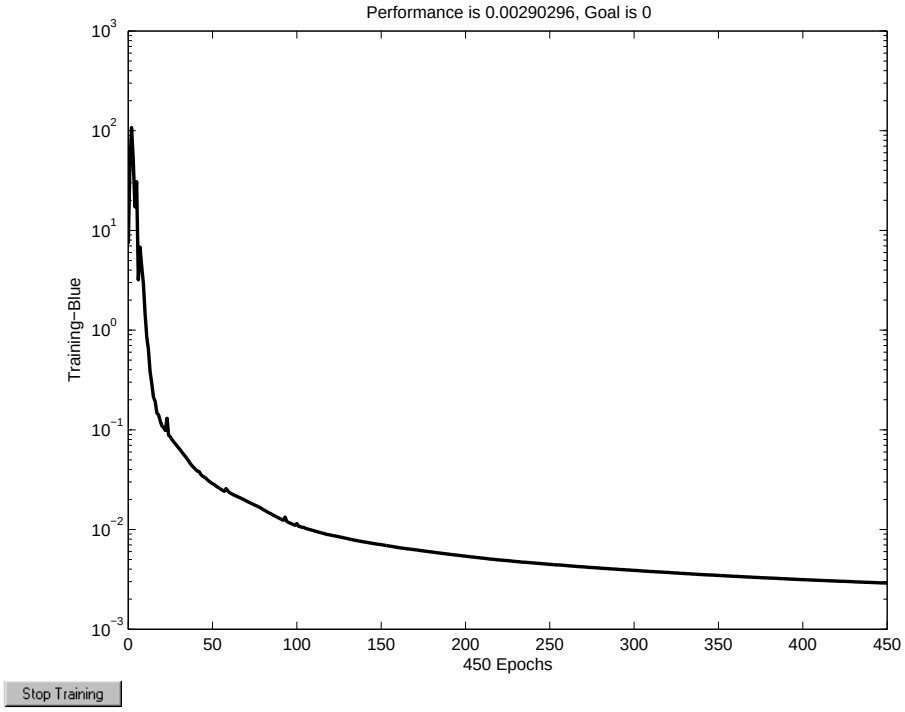
Training 4



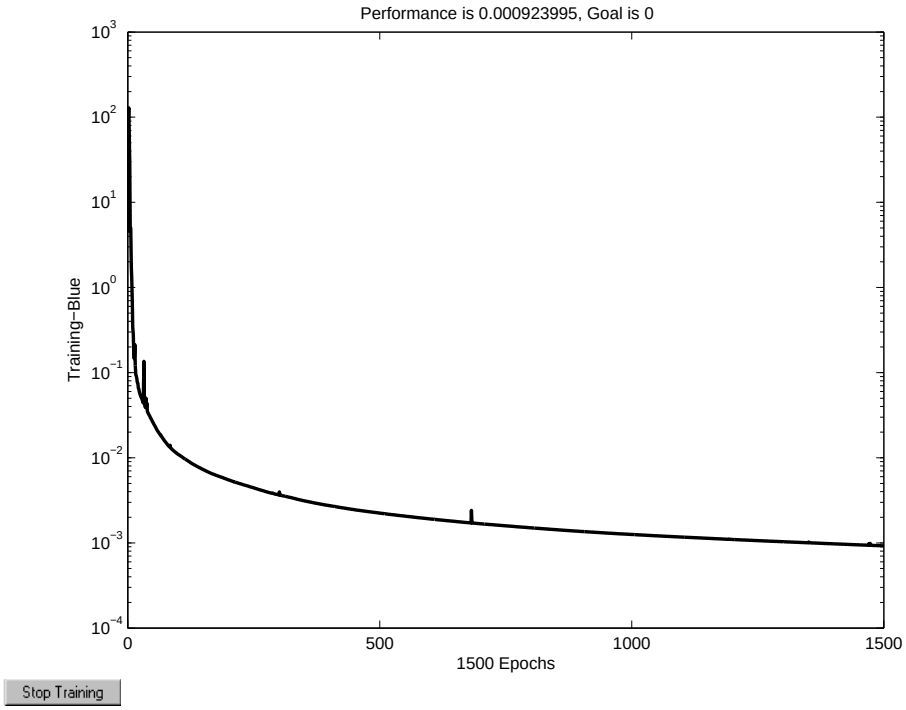
Training 5



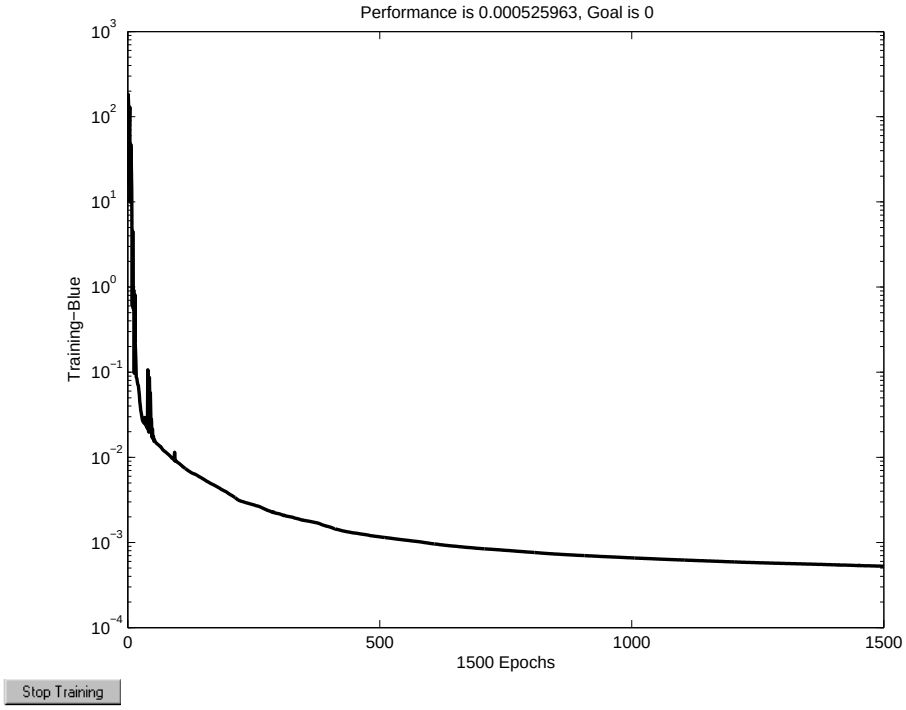
Training 6



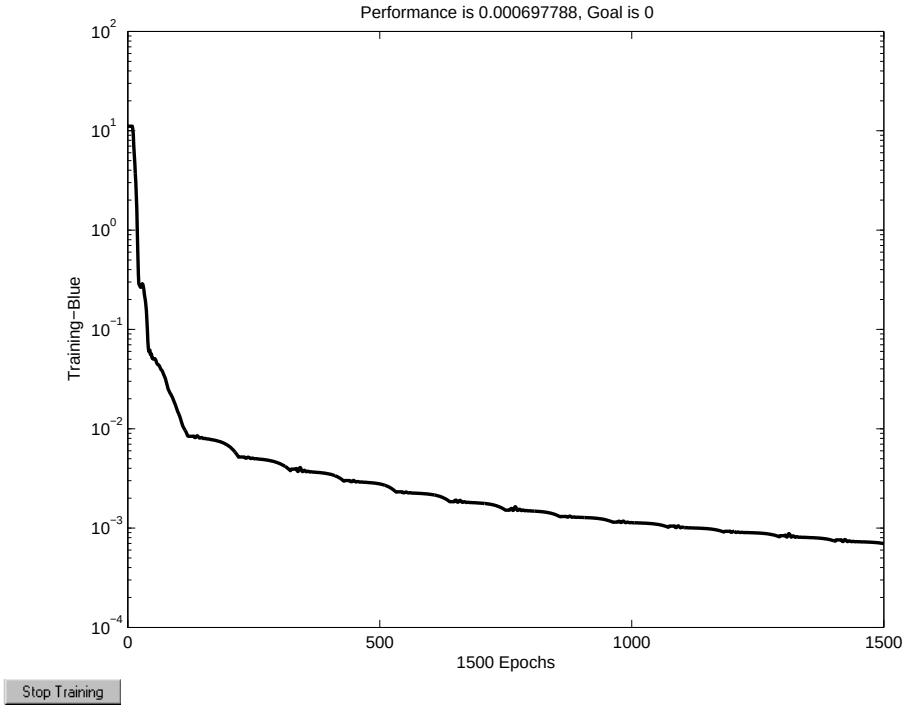
Training 7



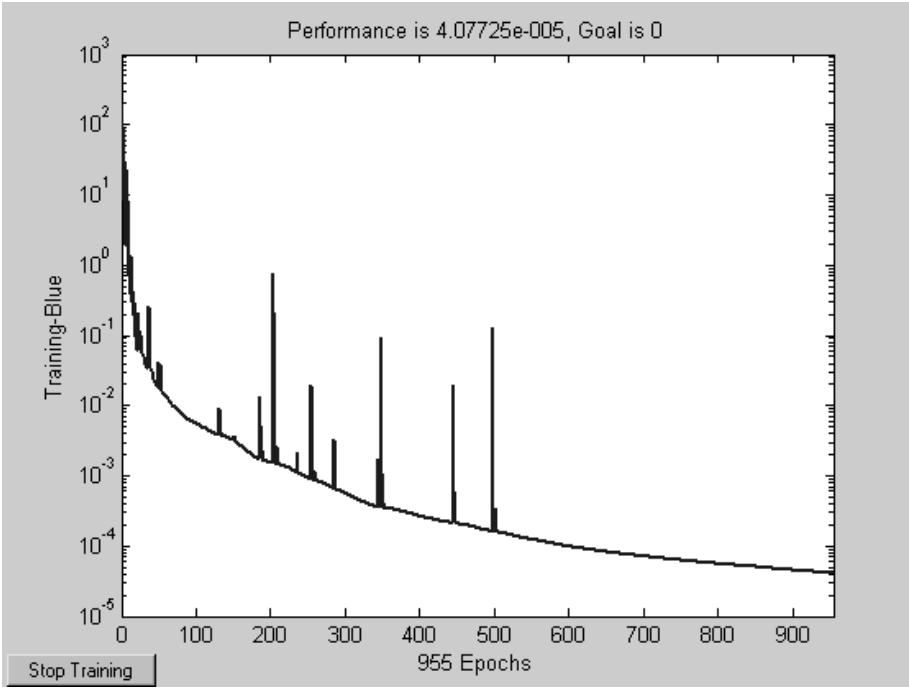
Training 8



Training 9



Training 10


Performance is 4.07725e-005, Goal is 0
Training-Blue
955 Epochs
Stop Training

Training 11

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